

SAIR

Spatial AI & Robotics Lab

CSE 473/573

L16: RECOGNITION & RETRIEVAL

Chen Wang

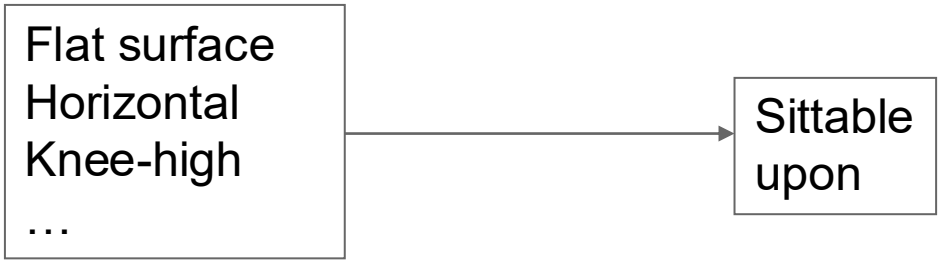
Spatial AI & Robotics Lab

Department of Computer Science and Engineering

University at Buffalo The State University of New York

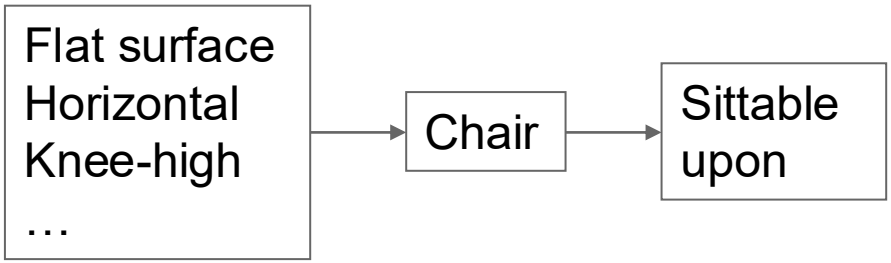
Why do we care about categories?

- Direct perception (affordances): [introduced by Gibson]

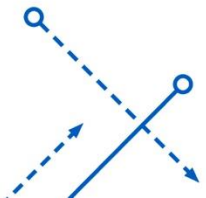




- Mediated perception (Categorization)



- Note:
  - Deciding that something is a chair might require access to more features than deciding that we can sit on something
  - A different level of categorization.



Limitations of Direct Perception

- Objects of similar structure might have very different functions



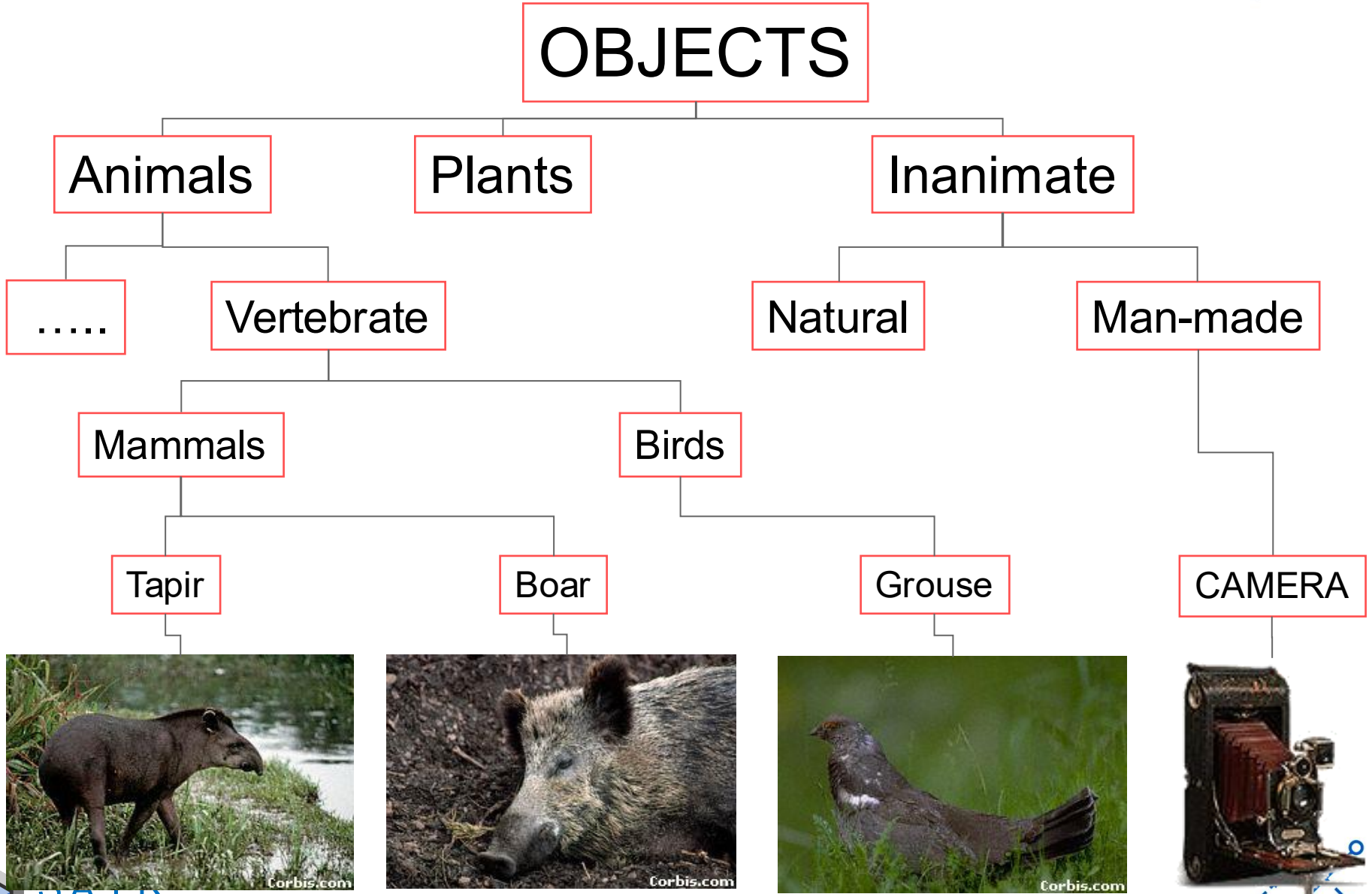
Figure 9.1.2 Objects with similar structure but different functions. Mailboxes afford letter mailing, whereas trash cans do not, even though they have many similar physical features, such as size, location, and presence of an opening large enough to insert letters and medium-sized packages.



Not all functions seem to be available from direct visual information only.

- The functions are the same at some level of description:
  - We can put things inside in both
  - Somebody will come later to empty them.
- However, we don't put inside the same kinds of things

Object Category



Challenges: robustness



Illumination





Object pose



Clutter



Occlusions



Intra-class appearance



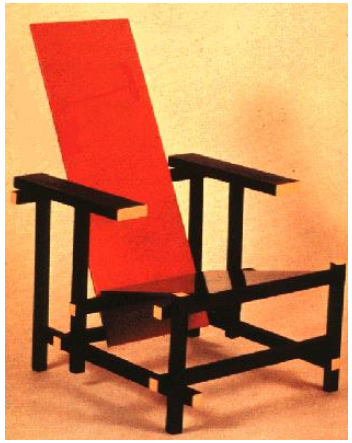
Viewpoint

Challenges: Within-class variations





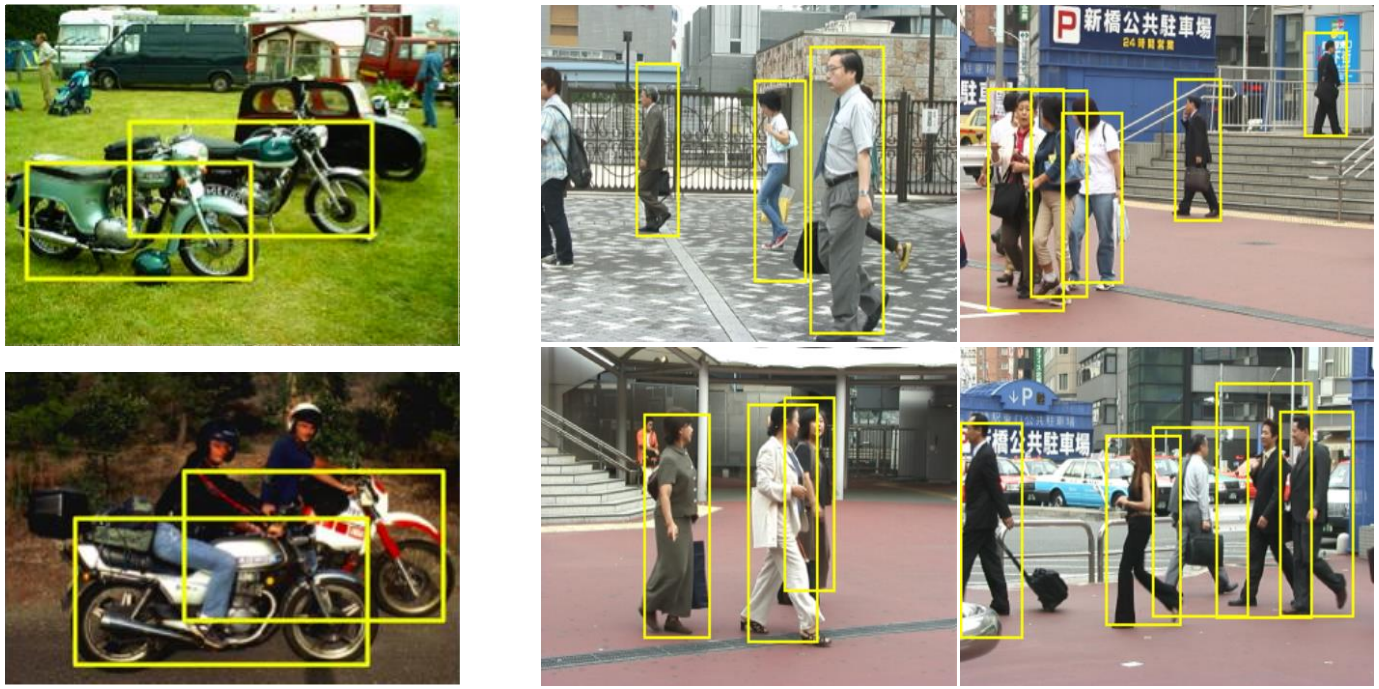








Challenges: Clutter, overlap, occlusion



Challenges: context and human experience



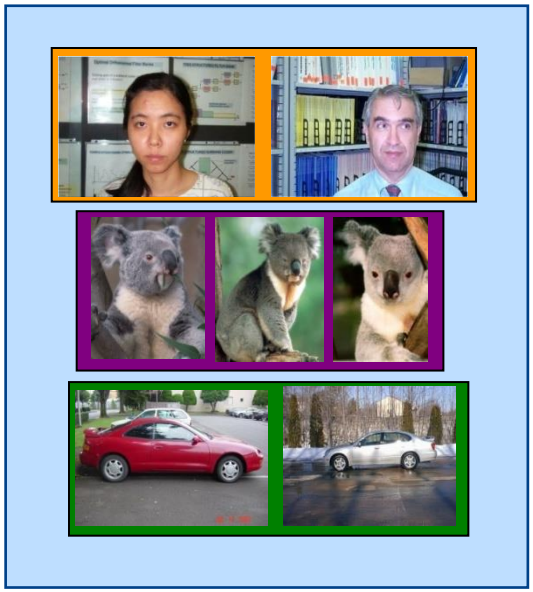


Challenges: learning with minimal supervision

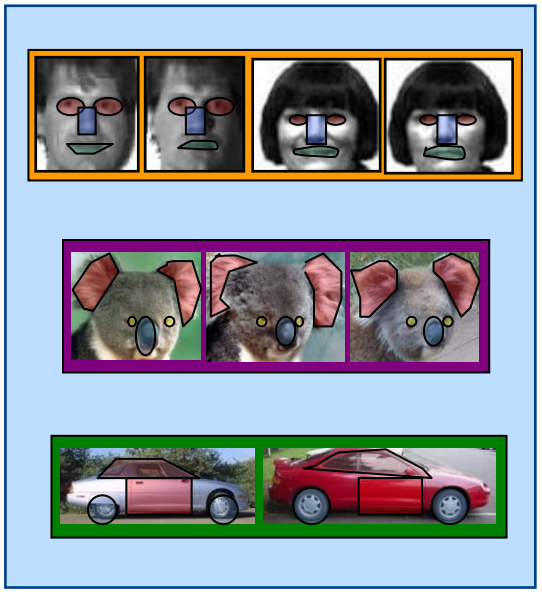
Less ⋮ ⋮ More



Unlabeled, multiple objects



Classes labeled, some clutter



Cropped to object, parts and classes labeled

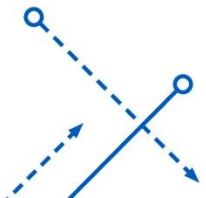
Inputs/outputs/assumptions

What is the **goal**?

- Say yes/no as to whether an object present in image

And/or:

- Categorize all objects
- Forced choice from pool of categories
- Bounding box on object
- Full segmentation
- Build a model of an object category
- Determine pose of an object, e.g., for robot to grasp



Bag-of-features models

- Still widely used in visual localization & mapping

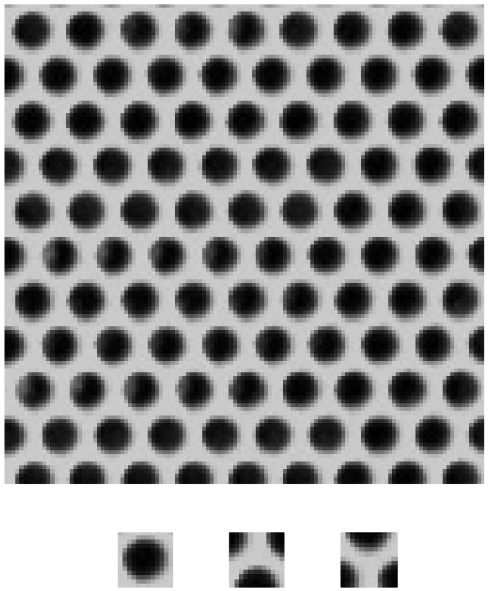
Represent an image as bag of features

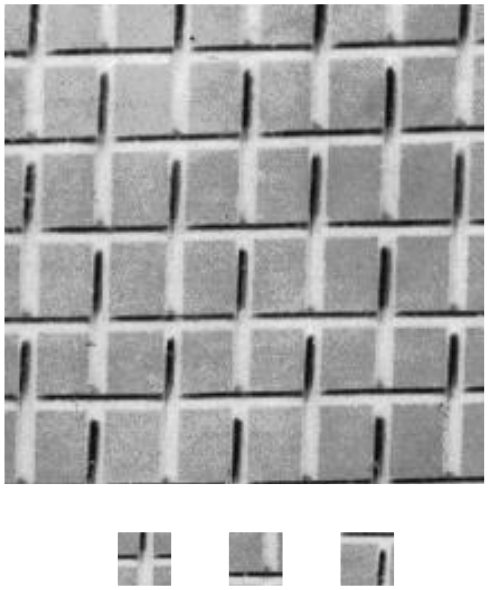


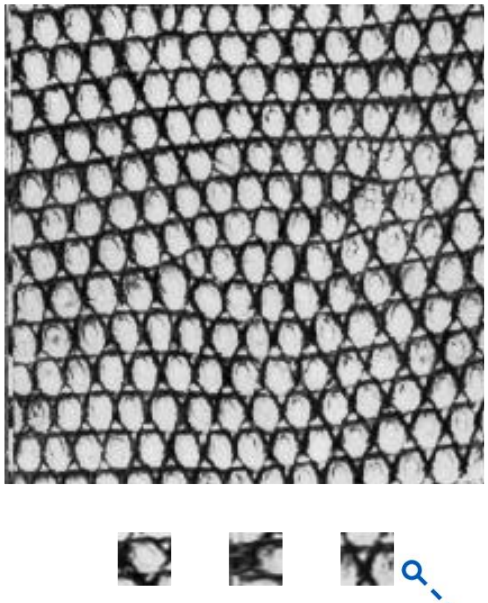


Origin 1: Texture recognition

- Texture is characterized by the **repetition** of **basic elements** or ***textons***
- For stochastic textures, the **identity** of the textons is more important than their **spatial** arrangement

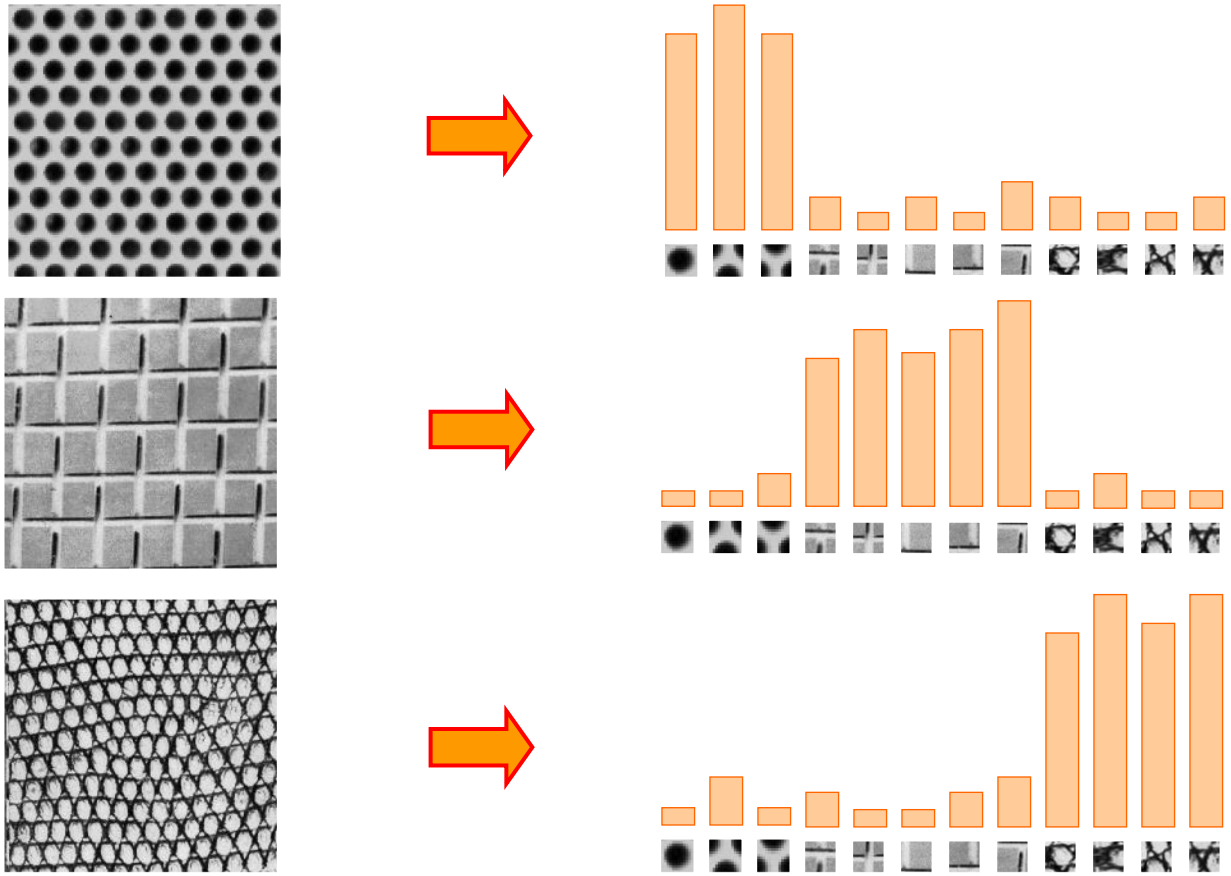






Julesz, 1981; Cula & Dana, 2001; Leung & Malik 2001; Mori, Belongie & Malik, 2001; Schmid 2001; Varma & Zisserman, 2002, 2003; Lazebnik, Schmid & Ponce, 2003

Origin 1: Texture recognition



Julesz, 1981; Cula & Dana, 2001; Leung & Malik 2001; Mori, Belongie & Malik, 2001; Schmid 2001; Varma & Zisserman, 2002, 2003; Lazebnik, Schmid & Ponce, 2003

Origin 2: Bag-of-words models

- Orderless document representation: frequencies of words from a dictionary Salton & McGill (1983)



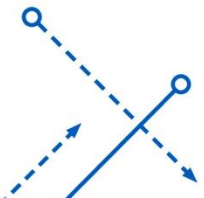
US Presidential Speeches Tag Cloud
http://chir.ag/phernalia/preztags/

Recognition

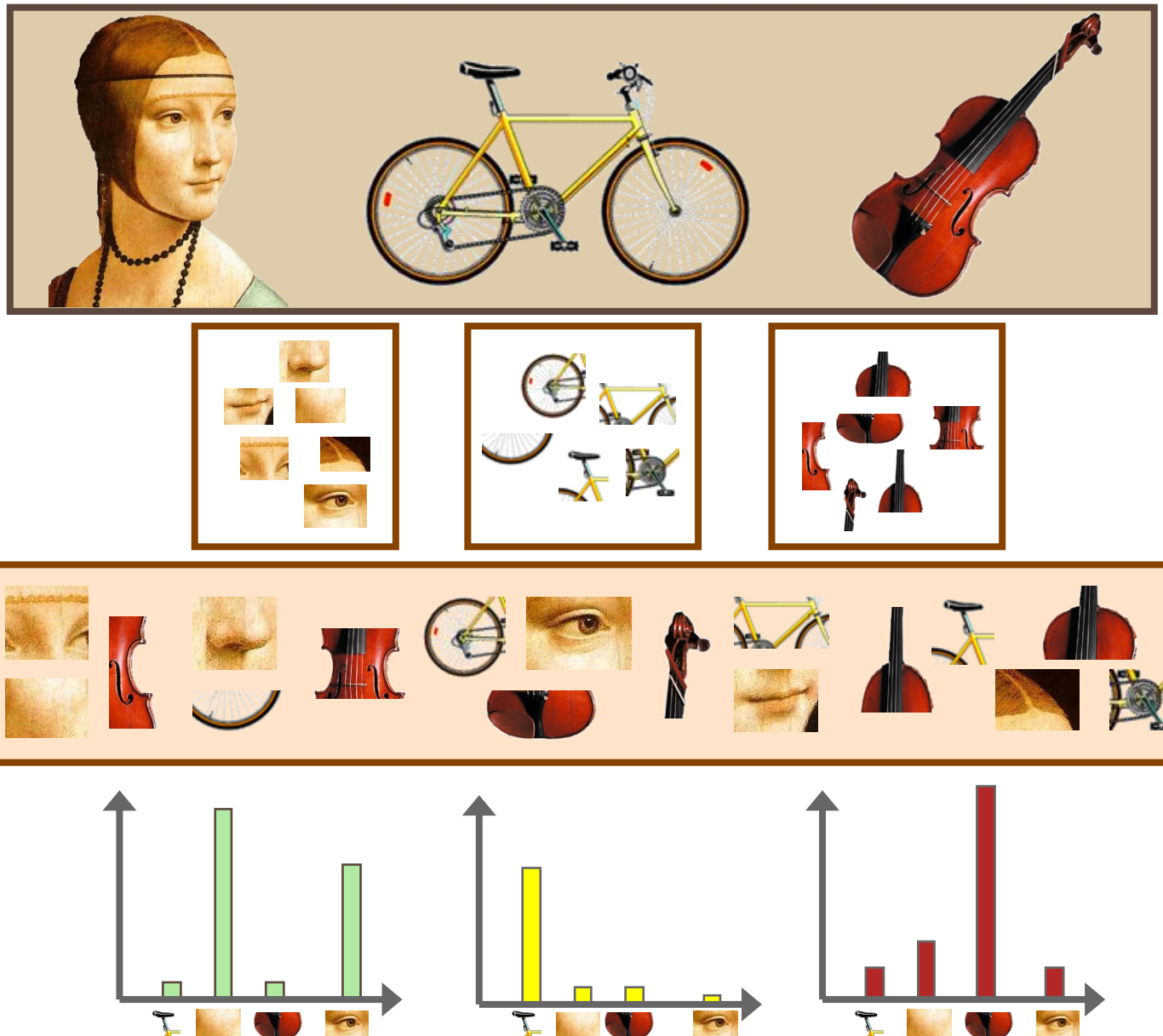
Bag-of-features steps (1 - 4)

1. Extract features
2. Learn “visual vocabulary”
3. Quantize features using visual vocabulary
4. Represent images by frequencies of “visual words”

Then predict the category using any classification models.

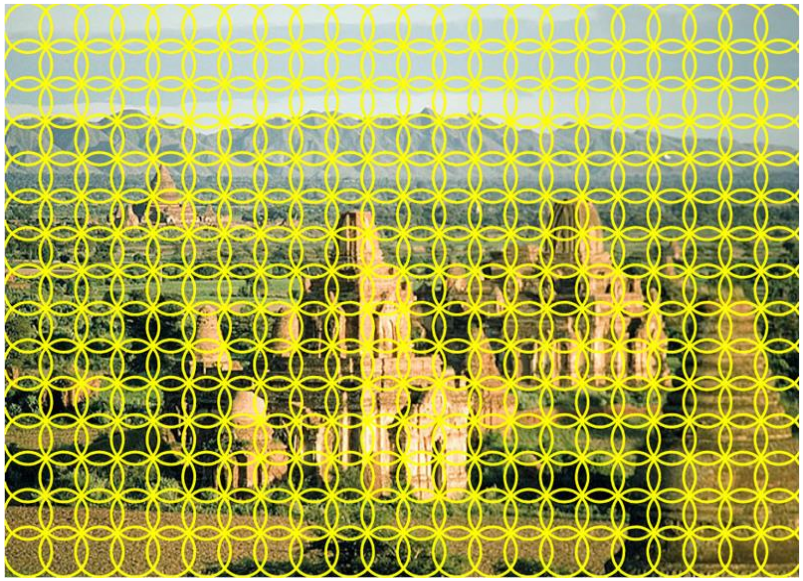


Bag-of-features steps



Feature extraction

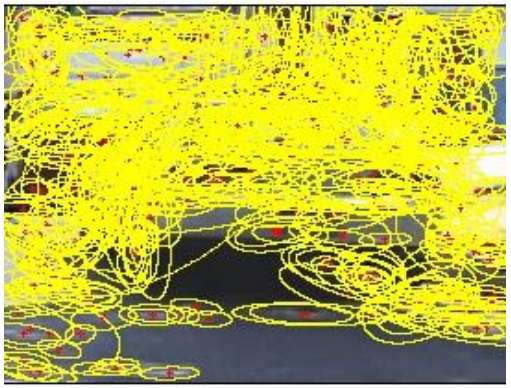
Regular grid



Interest regions



Sampling strategies



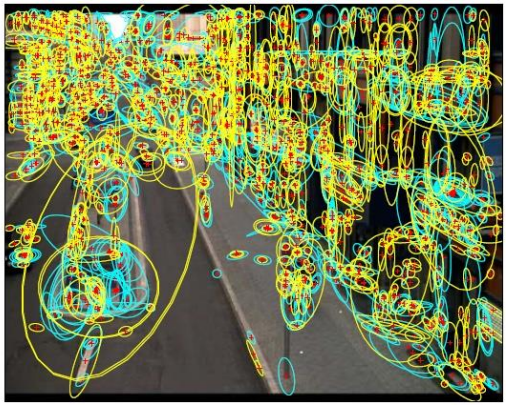
Sparse, at interest points



Dense, uniformly



Randomly



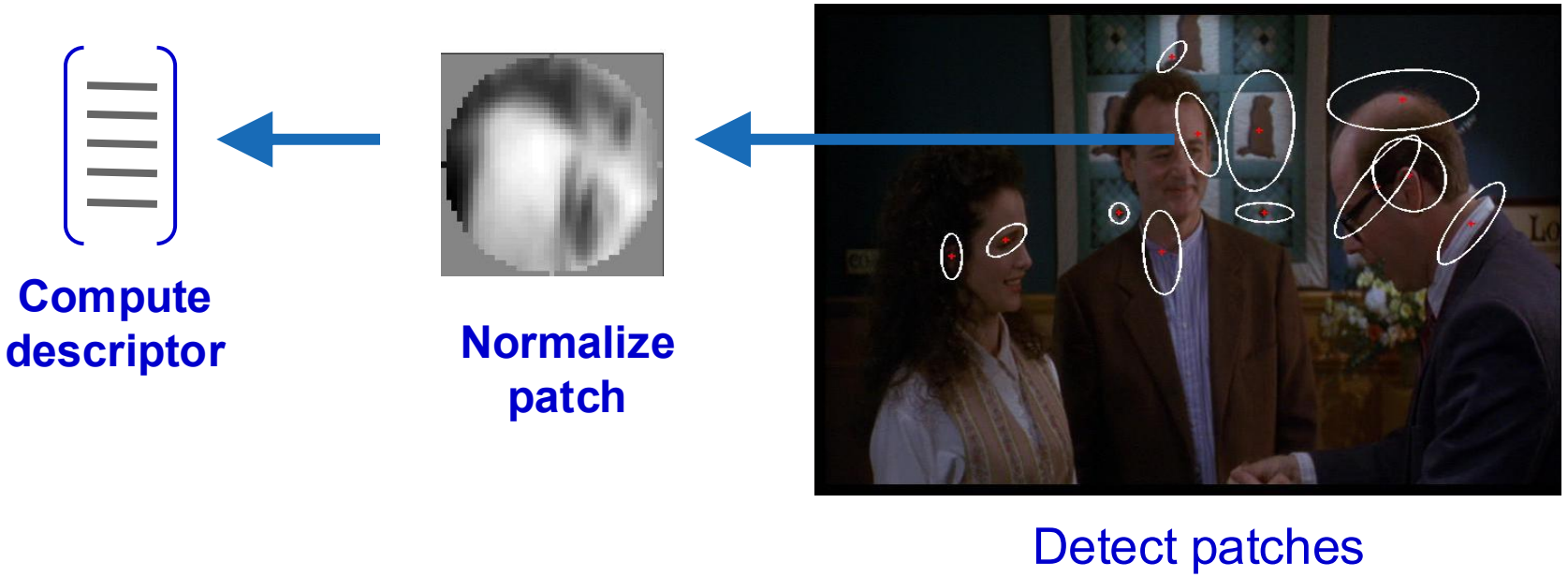
Multiple interest operators

- To find **specific**, textured **objects**, **sparse sampling** from interest points often more reliable.
- **Multiple** complementary interest **operators** offer more image coverage.
- For object **categorization**, **dense sampling** offers better coverage.

[See Nowak, Jurie & Triggs, ECCV 2006]

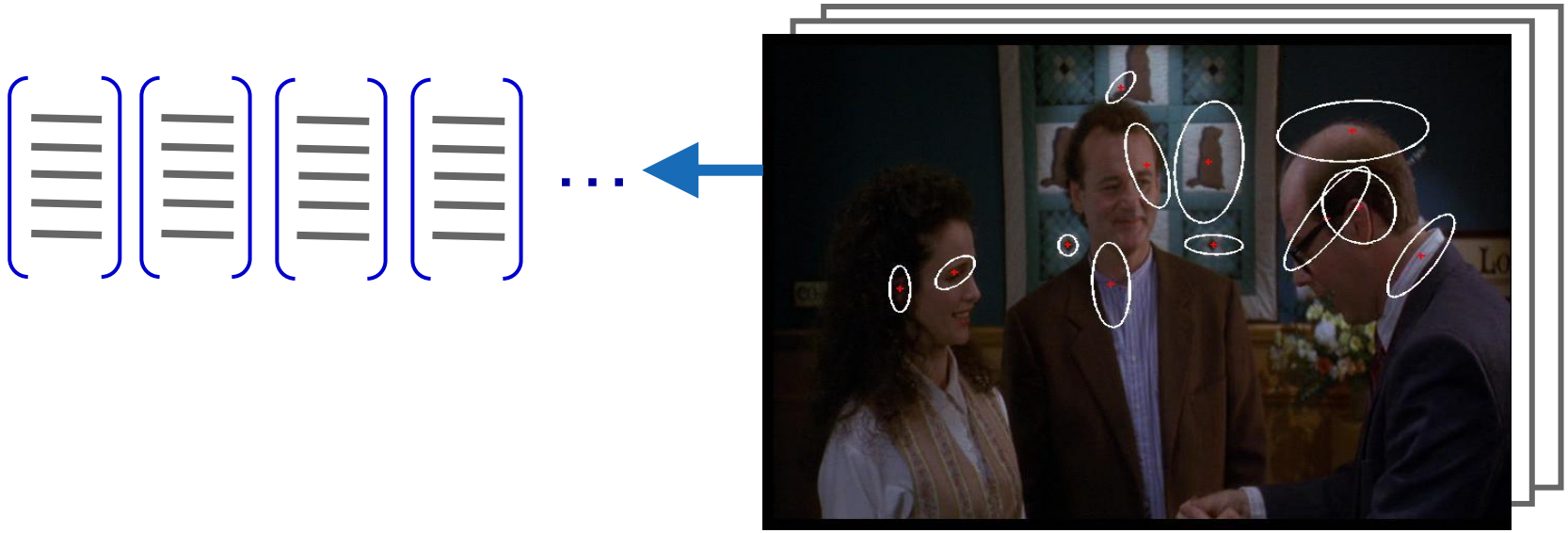
Image credits: F-F. Li, E. Nowak, J. Sivic

Feature extraction



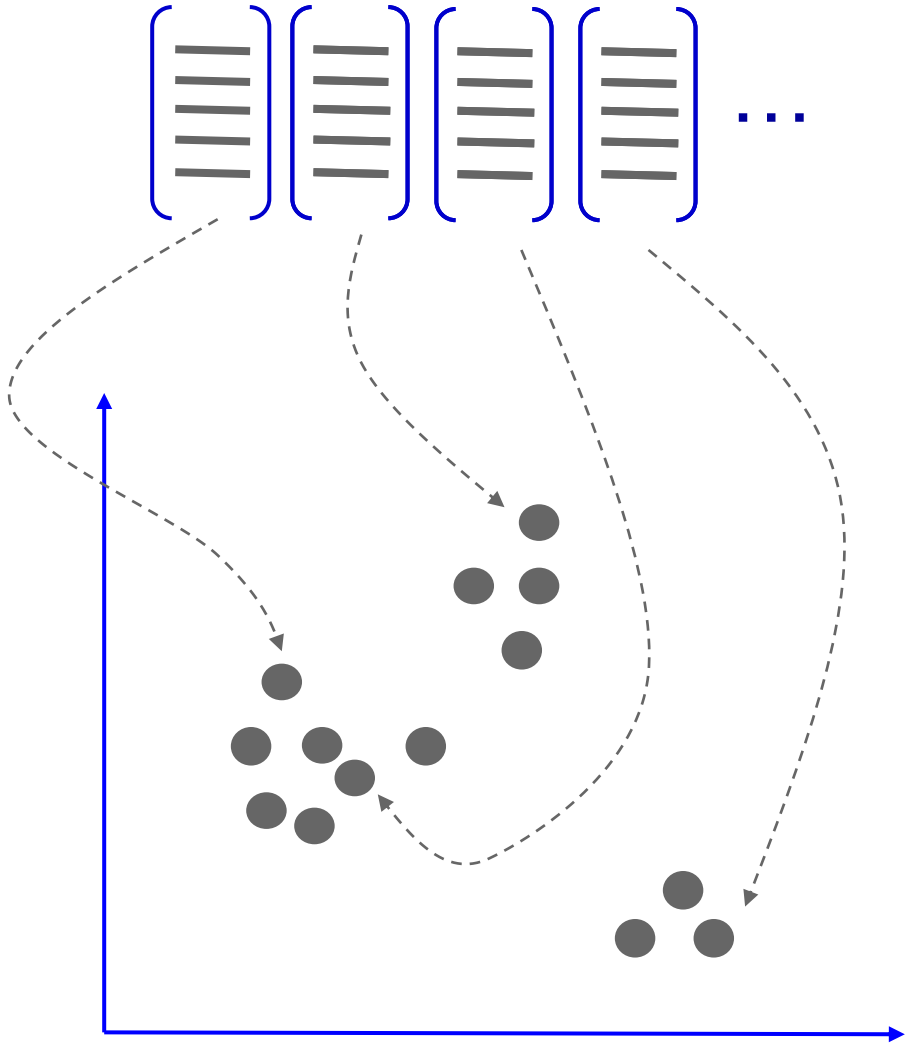
Slide credit: Josef Sivic

Feature extraction



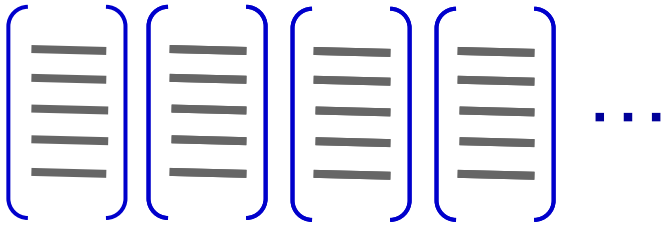
Slide credit: Josef Sivic

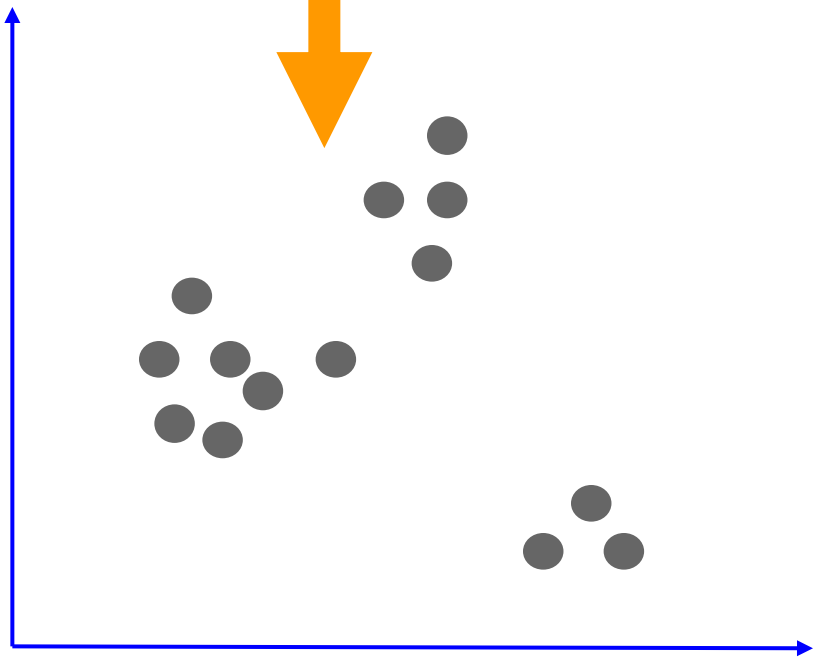
Learning the visual vocabulary

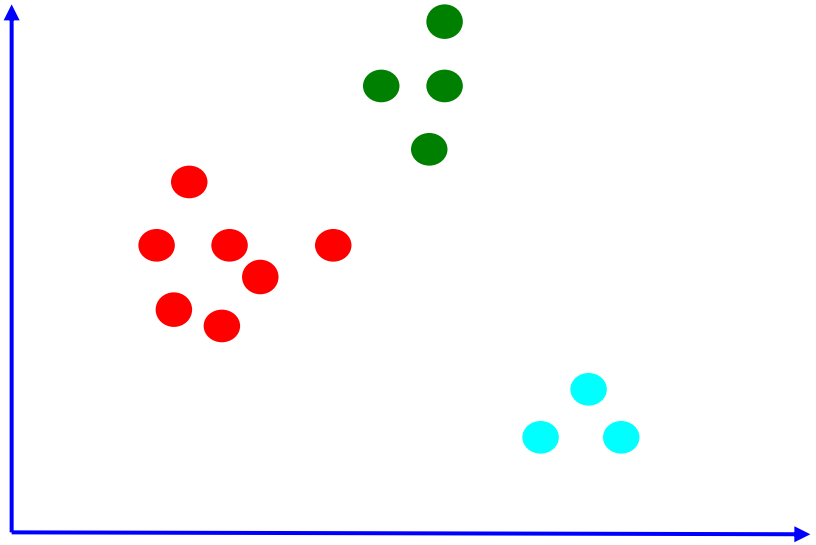


Slide credit: Josef Sivic

Learning the visual vocabulary



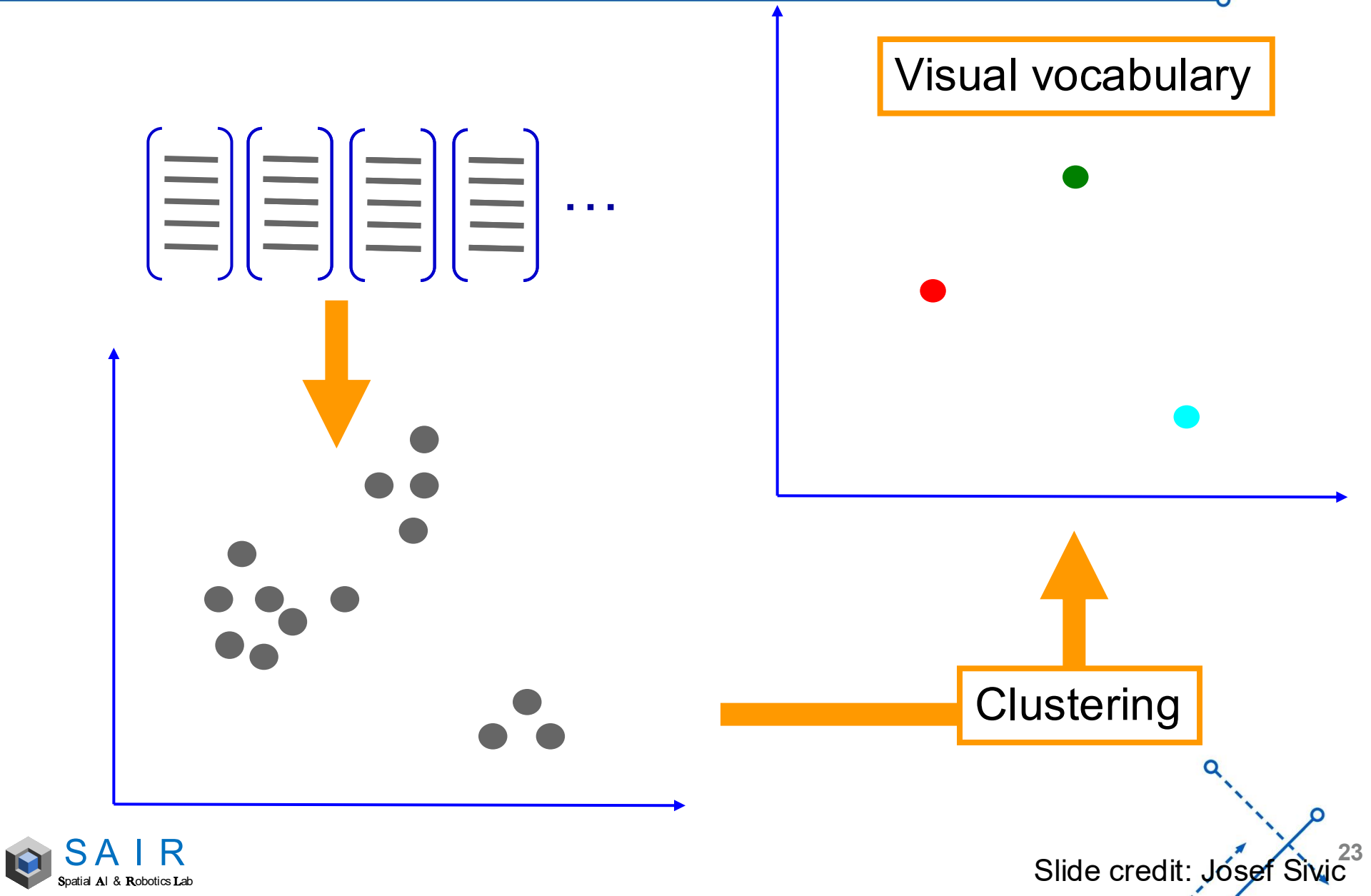




Clustering

Slide credit: Josef Sivic

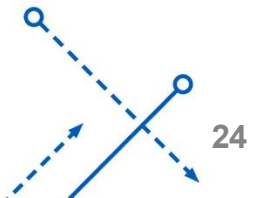
Learning the visual vocabulary



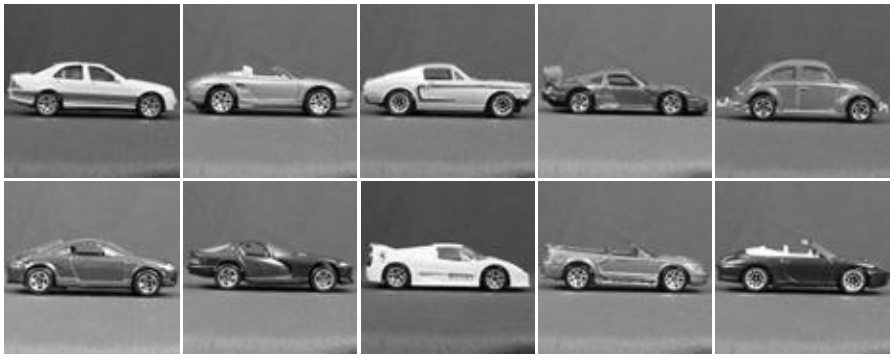
Slide credit: Josef Sivic

Clustering and vector quantization

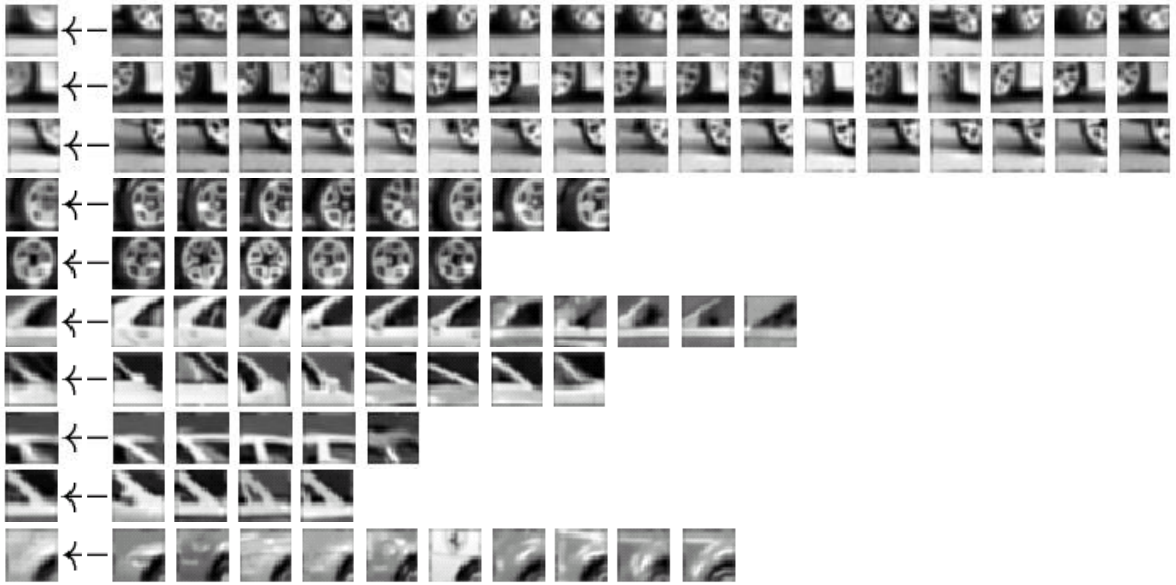
- Clustering: learning a visual vocabulary or codebook
  - Unsupervised learning process
  - Each cluster center (e.g., k-means) becomes a codevector
  - Codebook can be learned on a separate training set
  - If the training set is sufficiently representative, the codebook will be “universal”
- The codebook is used for quantizing features
  - A *vector quantizer* takes a feature vector and maps it to the index of the nearest codevector in a codebook
  - Codebook = visual vocabulary
  - Codevector = visual word



Example Codebook





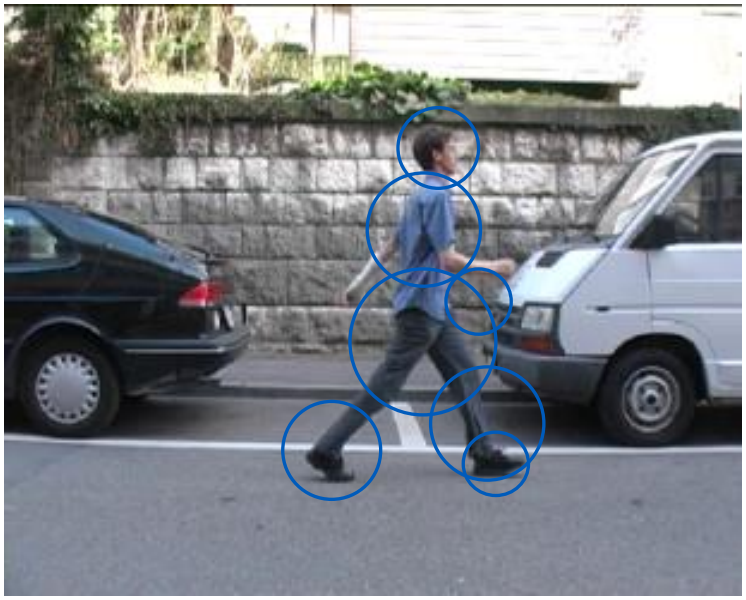


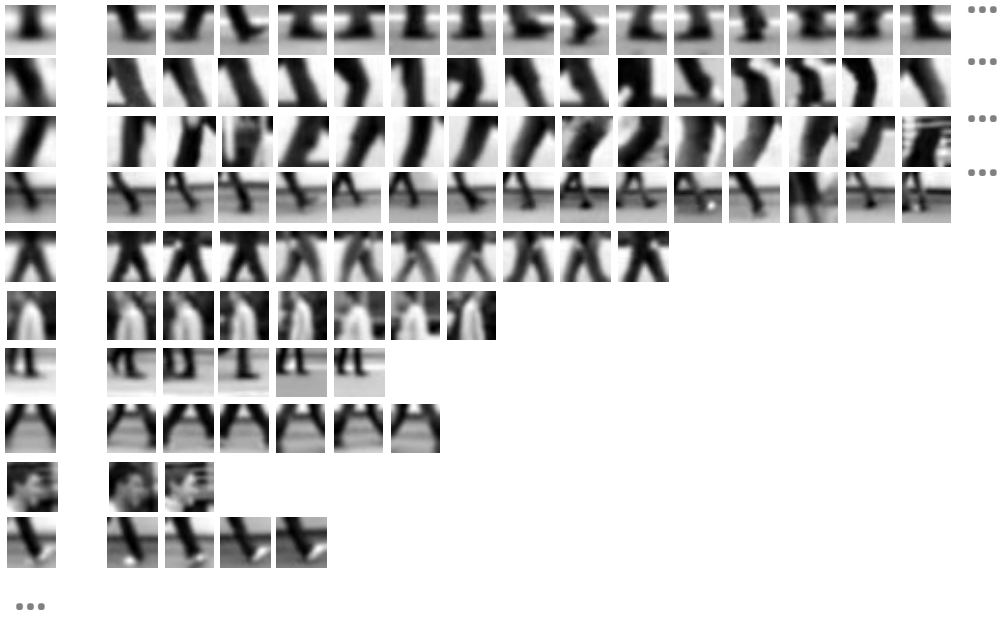
...

Appearance codebook

Source: B. Leibe

Another codebook





Appearance codebook

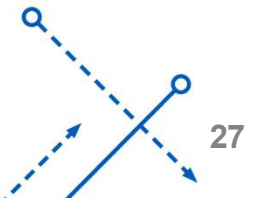
Source: B. Leibe

Recognition

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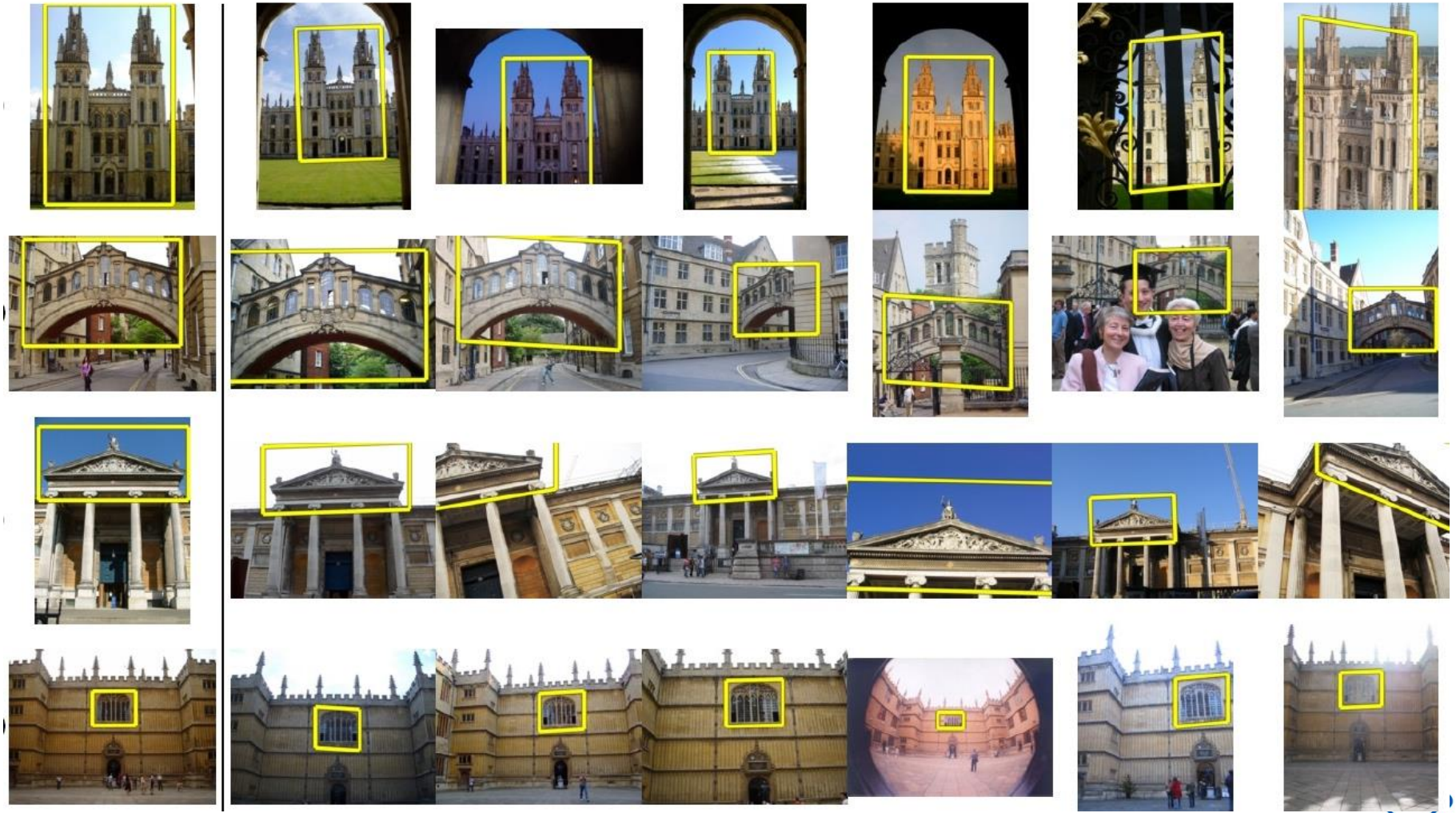
Then predict the category using any classification models.



Application: Image Retrieval

Query

Results from 5k Flickr images (demo available for 100k set)



[Philbin CVPR'07]

Local Retrieval



Query region



Retrieved frames

Efficient Retrieval

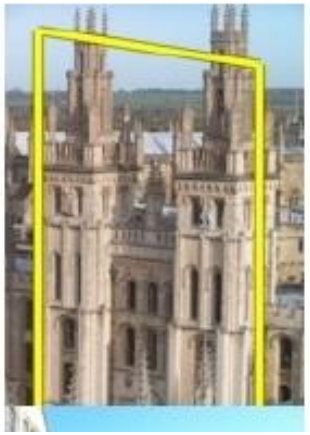
How to quickly find images in a large database that match a given image region?



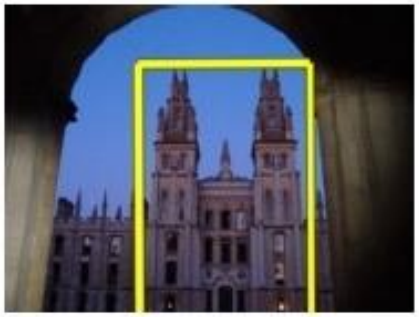








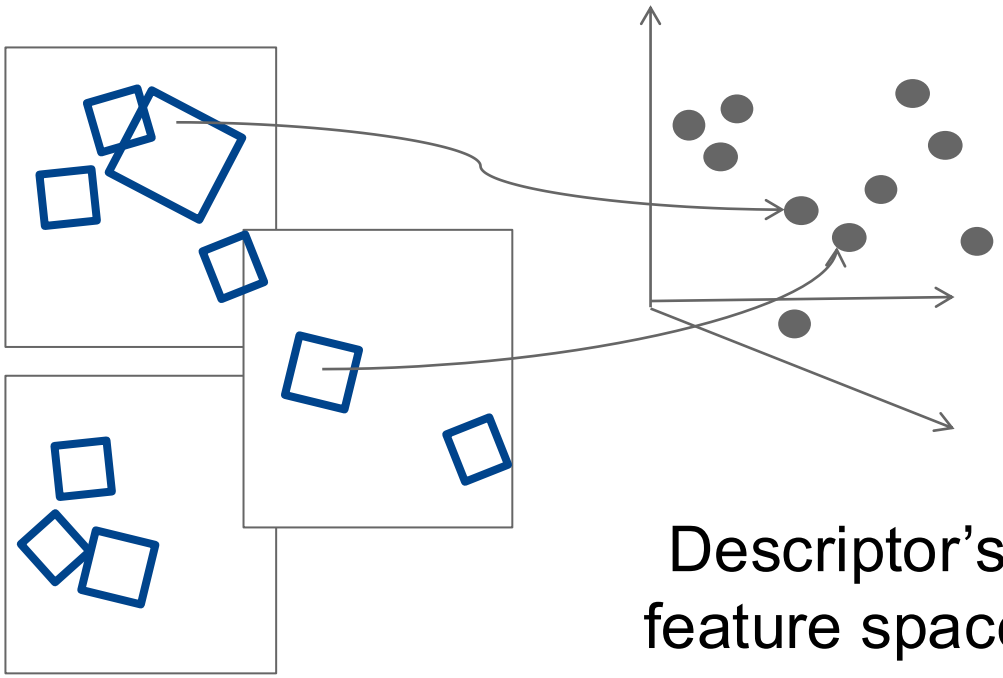






Indexing local features

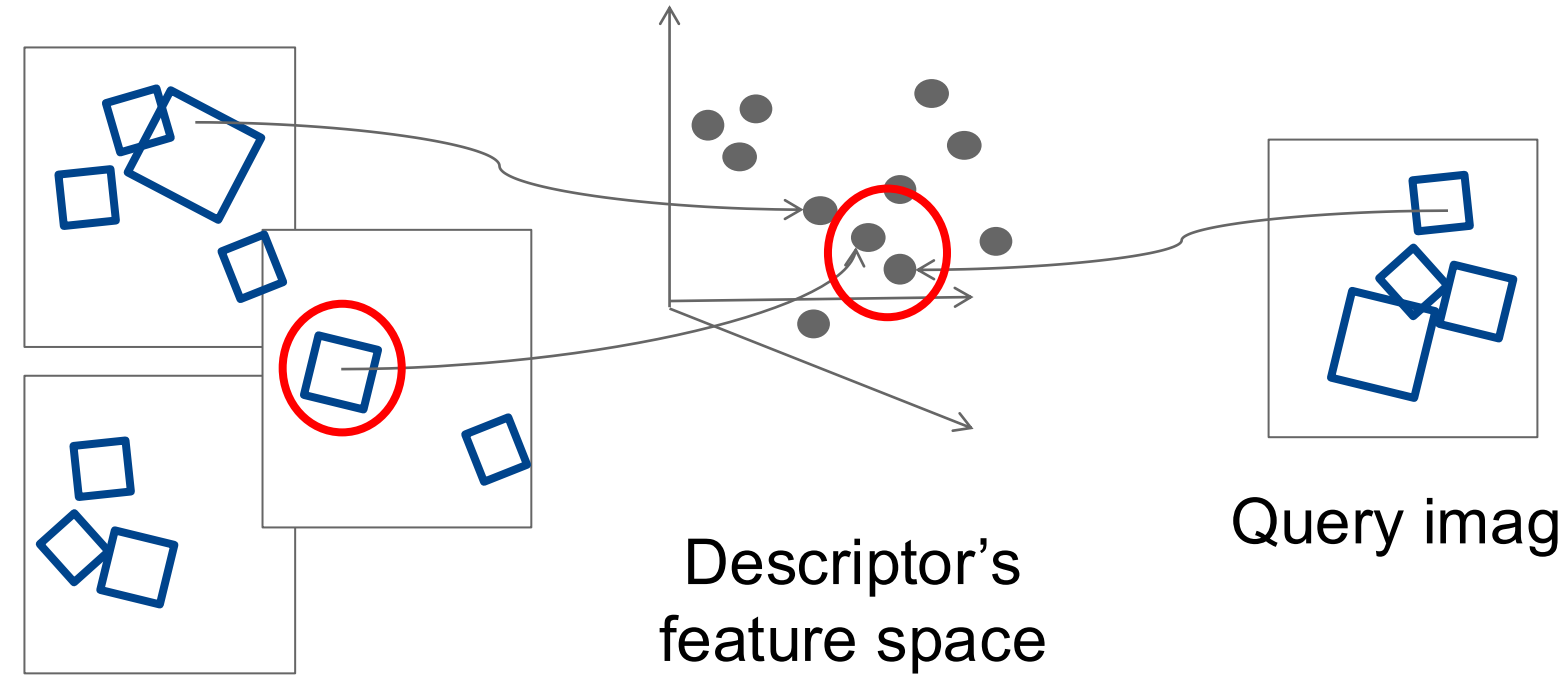
- Each patch / region has a **descriptor**, which **is a point** in some high-dimensional feature space, e.g., SIFT.



Descriptor's feature space

Indexing local features

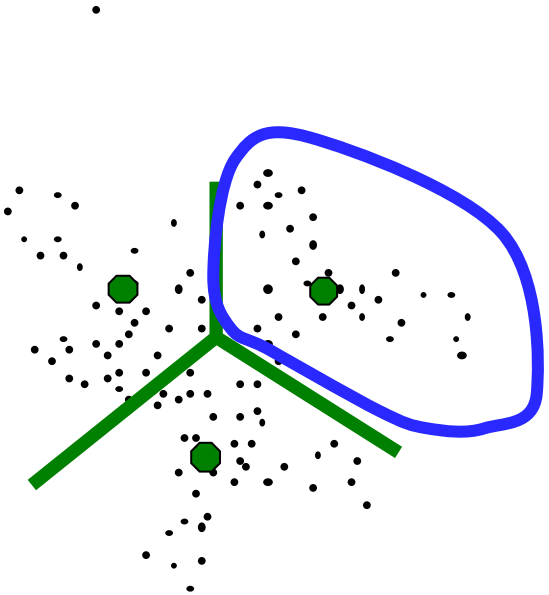
- When we see close points in feature space, we have similar descriptors, which indicates similar local content.



Easily have millions of features to search!

Visual words

- Example: each group of patches belongs to the same visual word



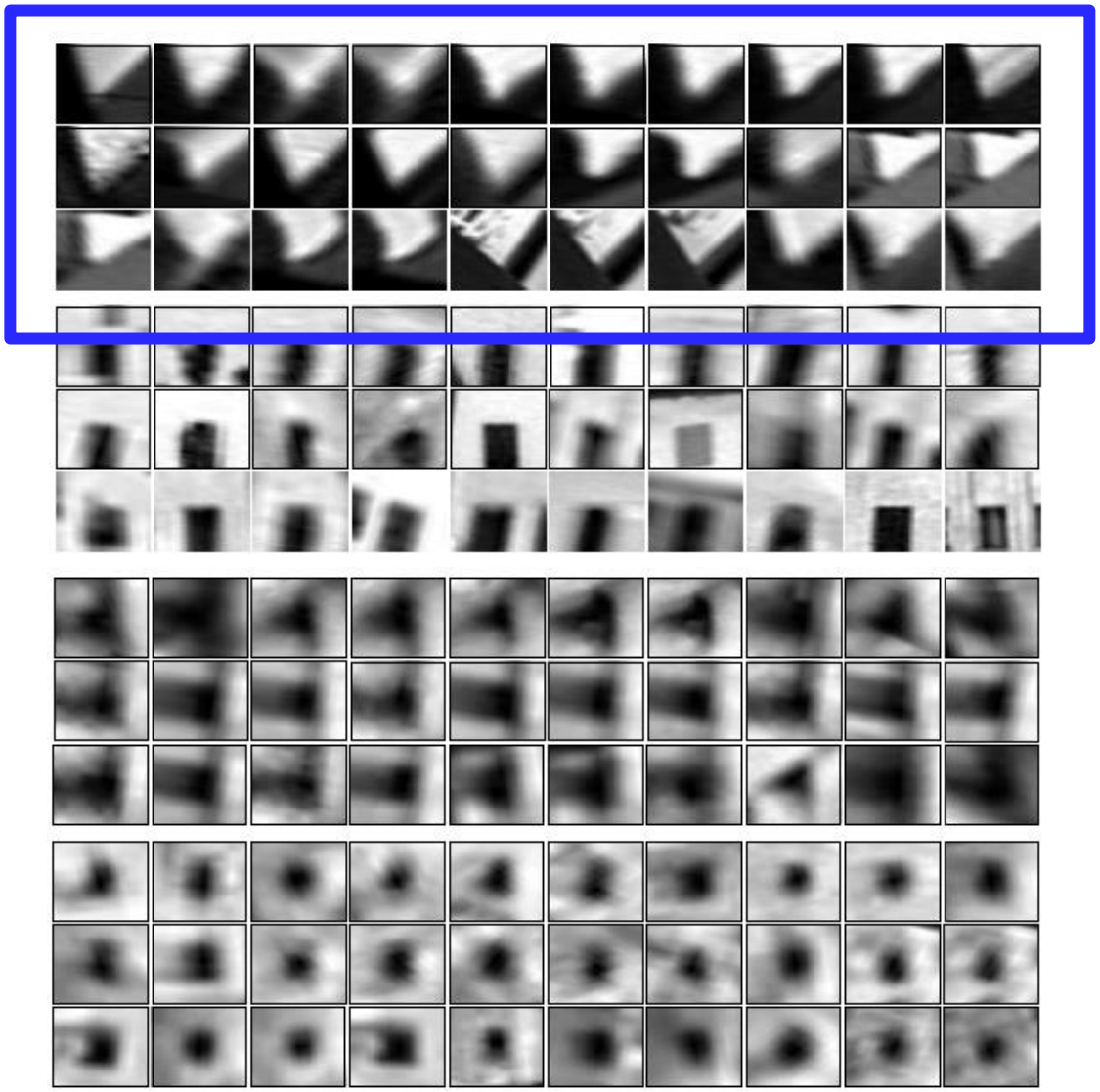


Figure from Sivic & Zisserman, ICCV 2003

Kristen Grauman

Indexing local features: inverted file index

- For text documents, an efficient way to find all *pages* on which a *word* occurs is to use an index…
- We want to find all *images* in which a *feature* occurs.
- To use this idea, we map features to "**visual words**".



Index

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"Drive I-95," From Boston to Florida; *inside back cover*
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Inverted file index

- Database images are loaded into the index mapping words to image numbers



Image #1

Image #2

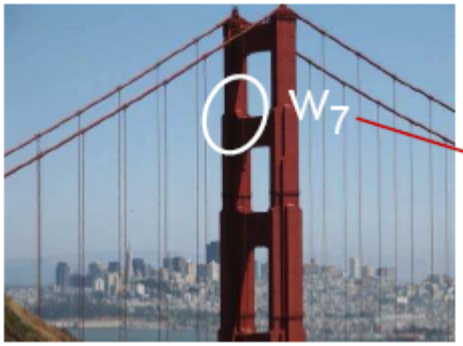
Image #3

Word #	Image #
1	3
2 …	
7	1, 2
8	3
9	
10 …	
91	2

…

Inverted file index

- New query image is mapped to indices of database images that share a word.



New query image

Word #	Image #
1	3
2	
…	
7	1, 2
8	3
9	
10	
…	
91	2



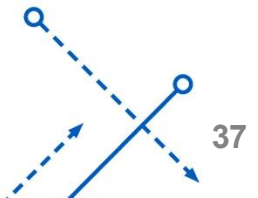
Image #1



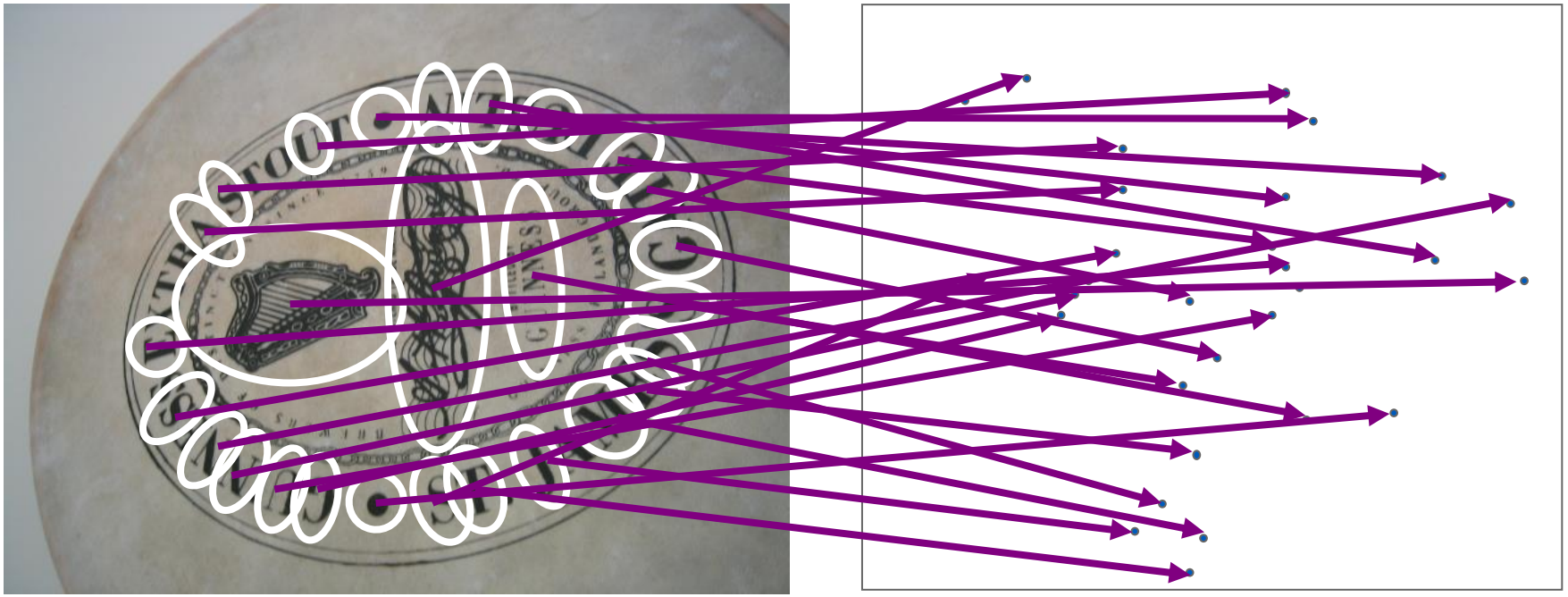
Image #2

Inverted file index

- Key requirement for inverted file index to be efficient:
  - Sparsity
    - If most pages/images contain most words, then it's no better than exhaustive search.
    - Comparing the words of a query versus every page.
  - Computational efficiency:
    - Idea: Vocabulary K-d tree:

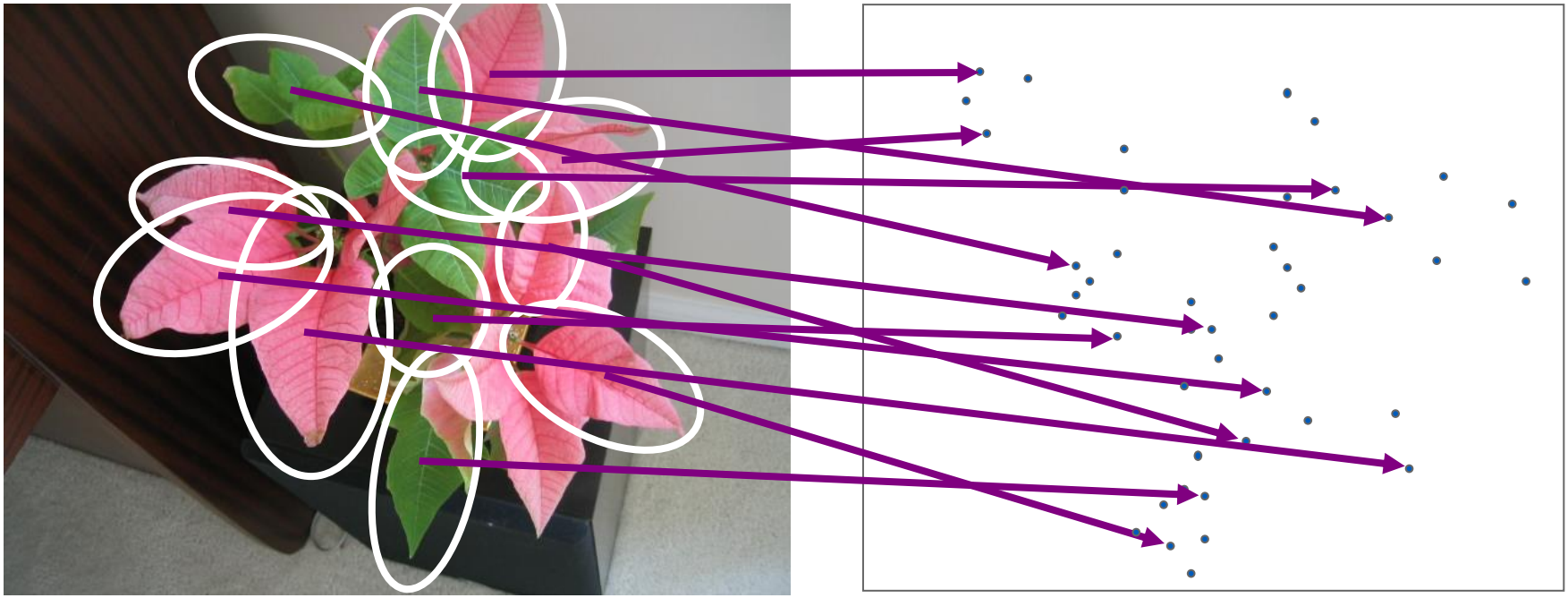


Recognition with K-d Tree



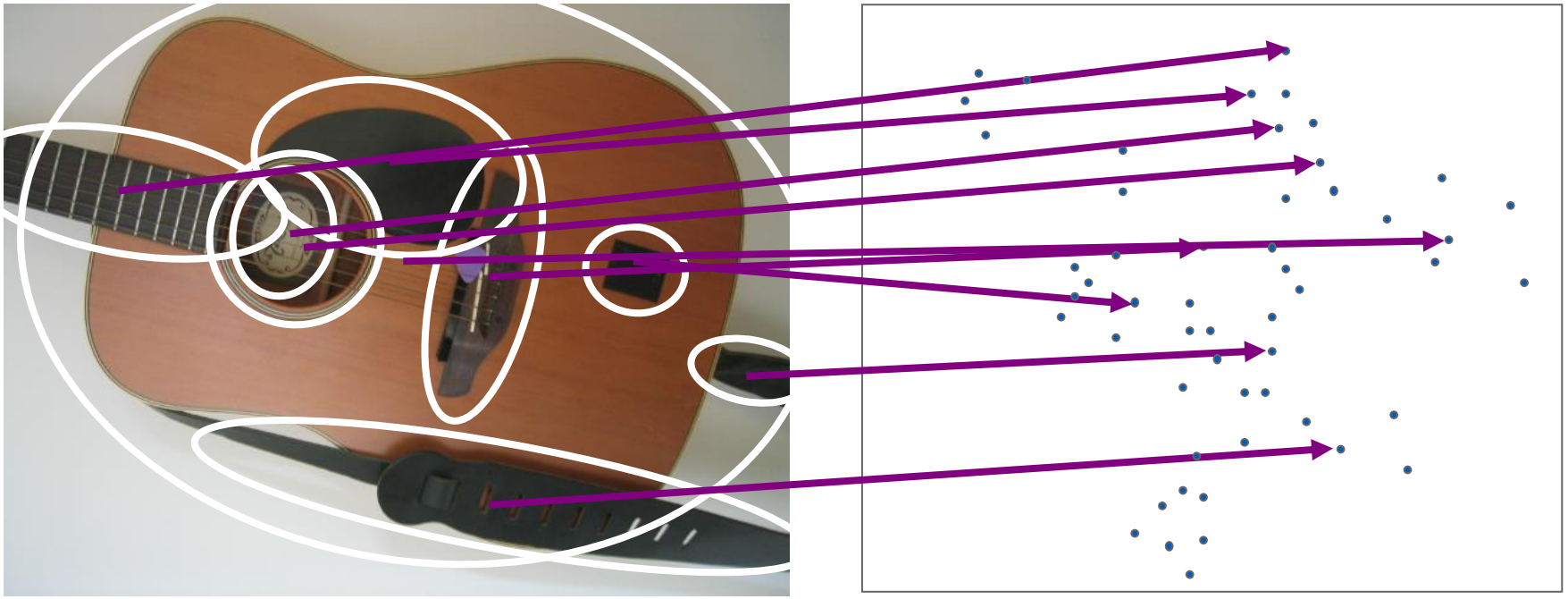
Following slides by David Nister (CVPR 2006)

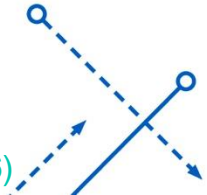
Recognition with K-d Tree



Following slides by David Nister (CVPR 2006)

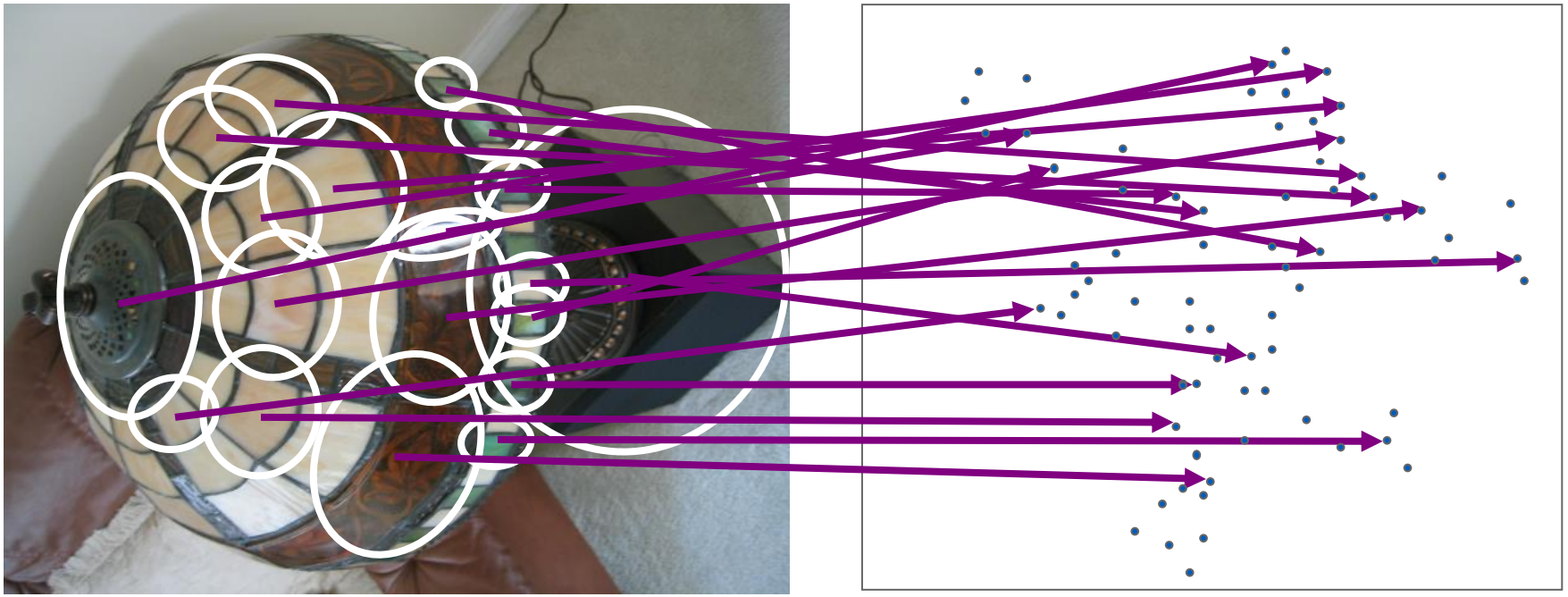
Recognition with K-d Tree





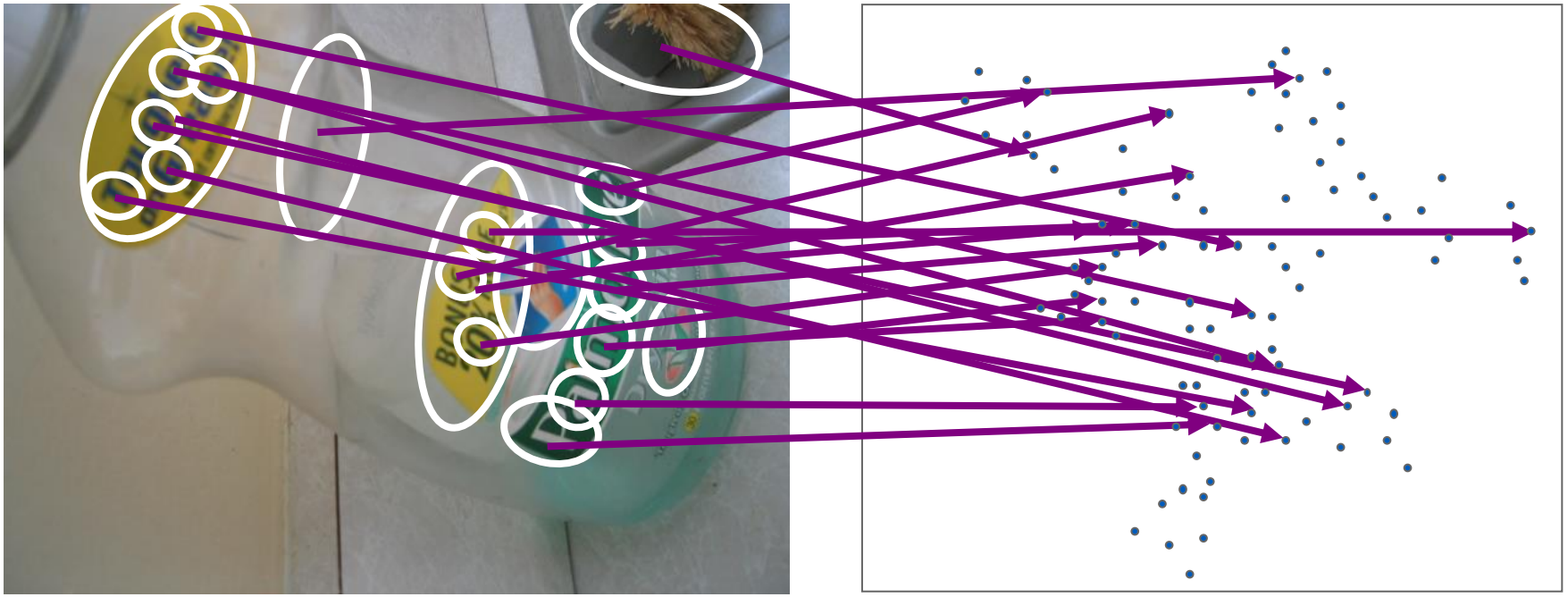
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



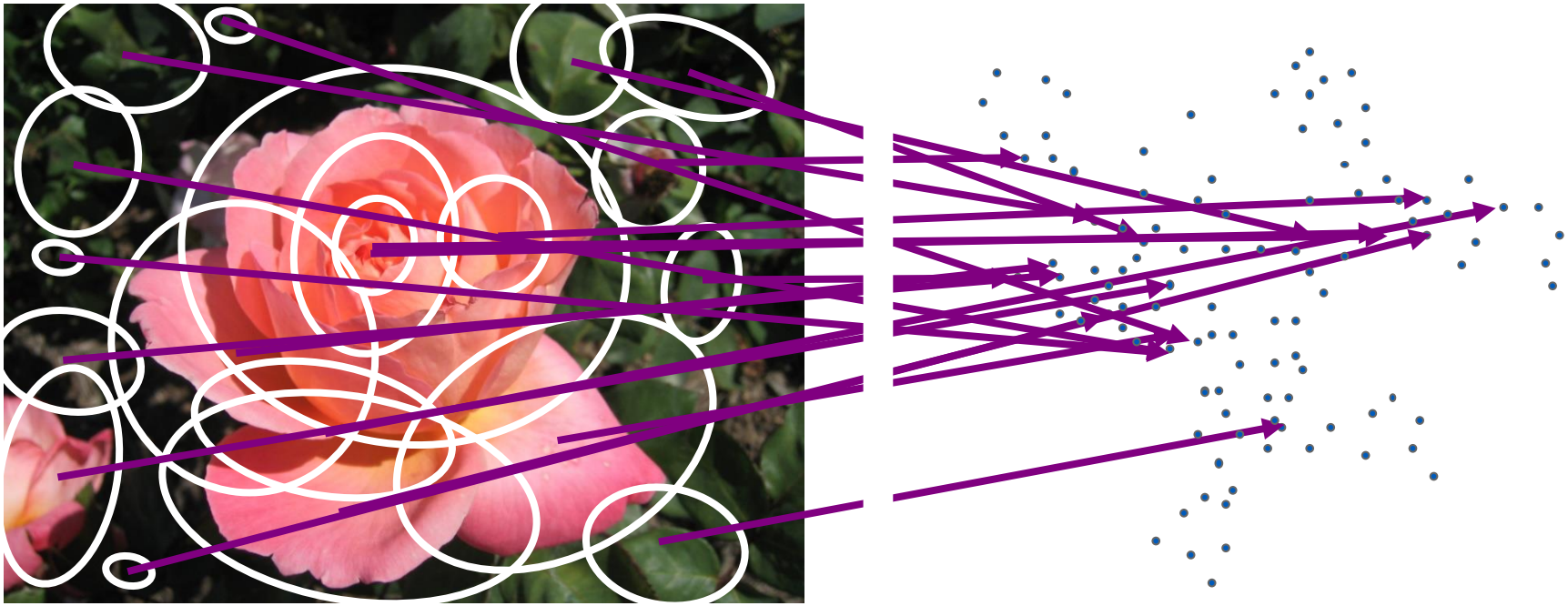
Following slides by David Nister (CVPR 2006)

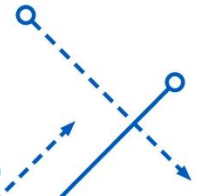
Recognition with K-d Tree



Following slides by David Nister (CVPR 2006)

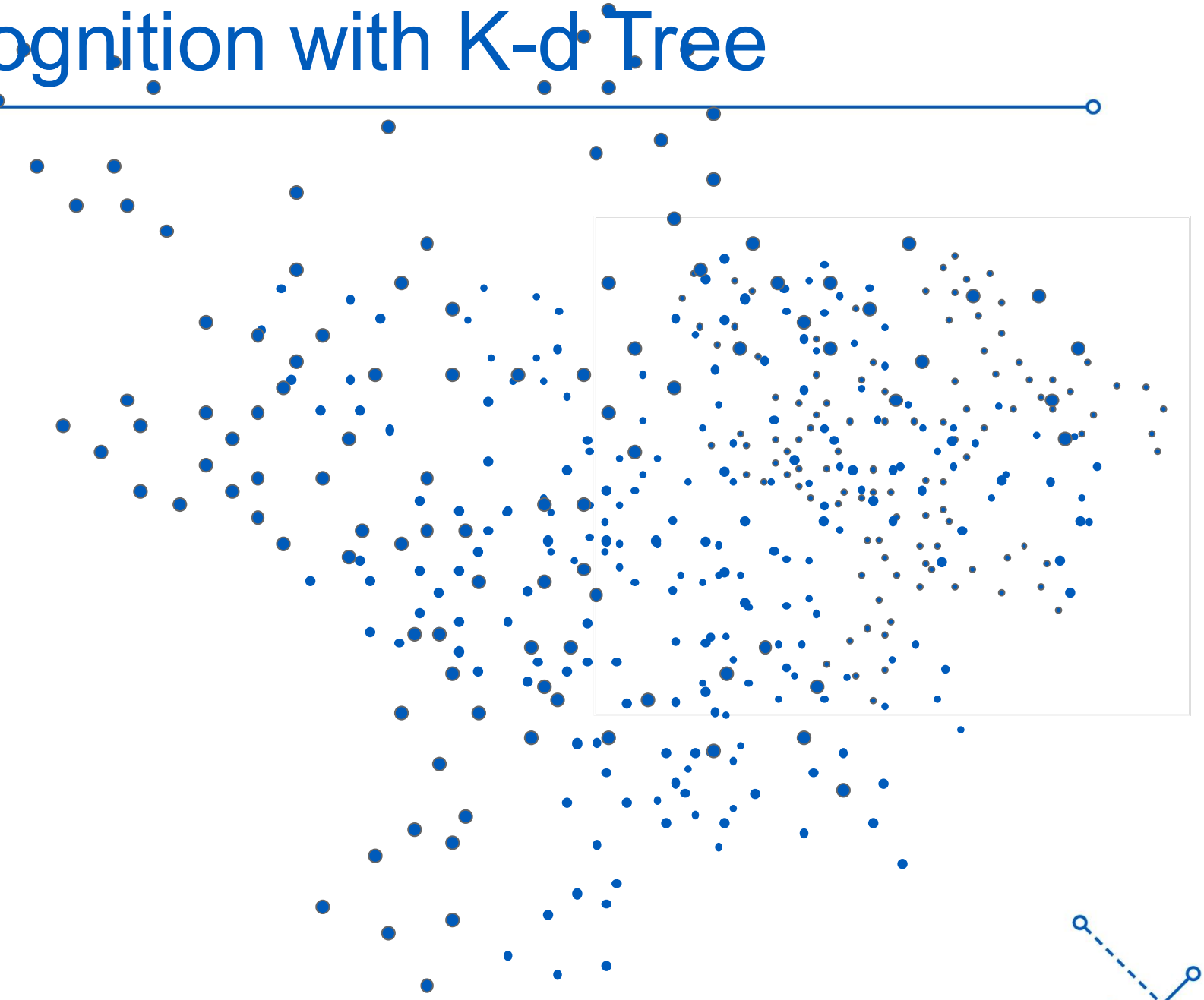
Recognition with K-d Tree





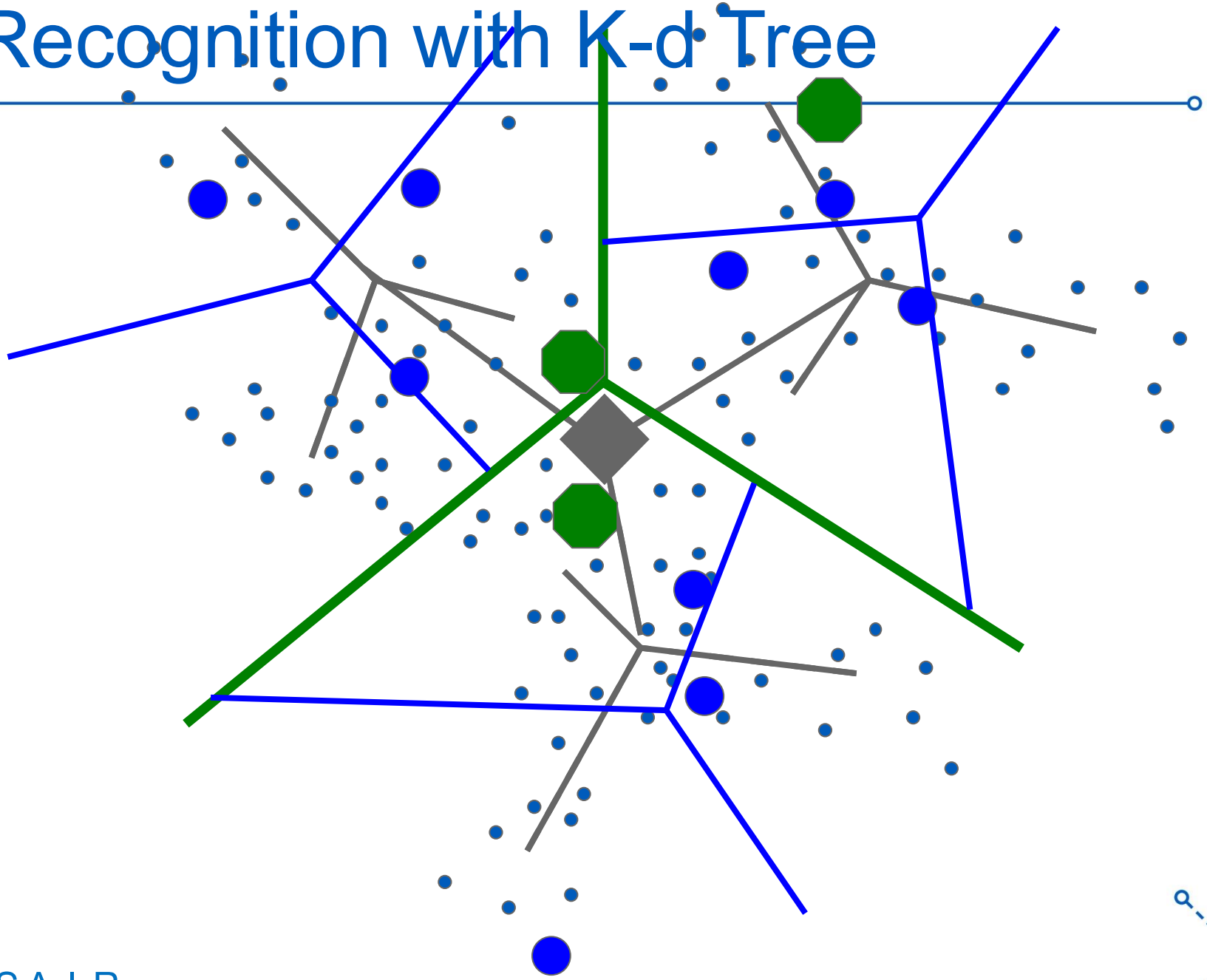
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



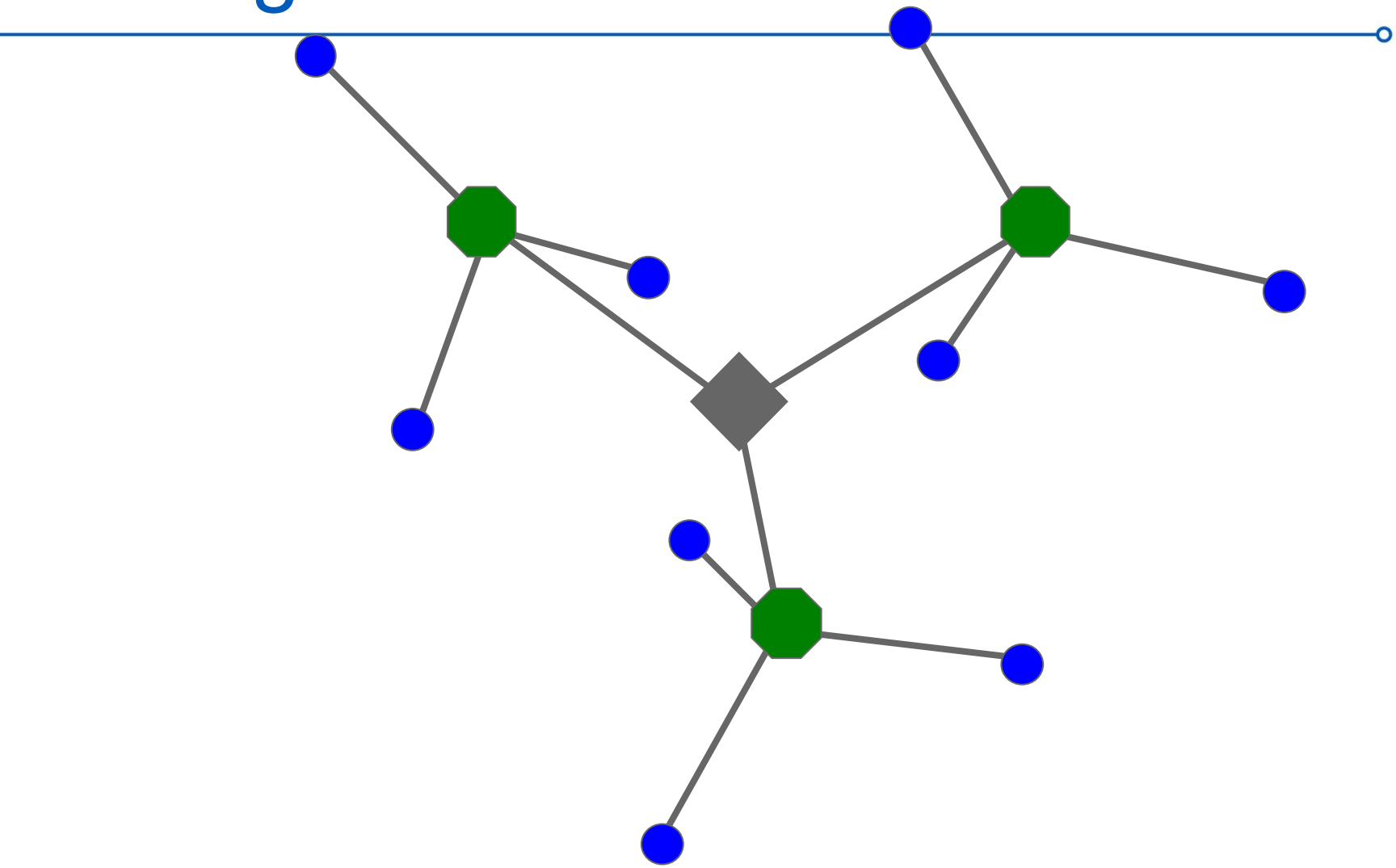
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



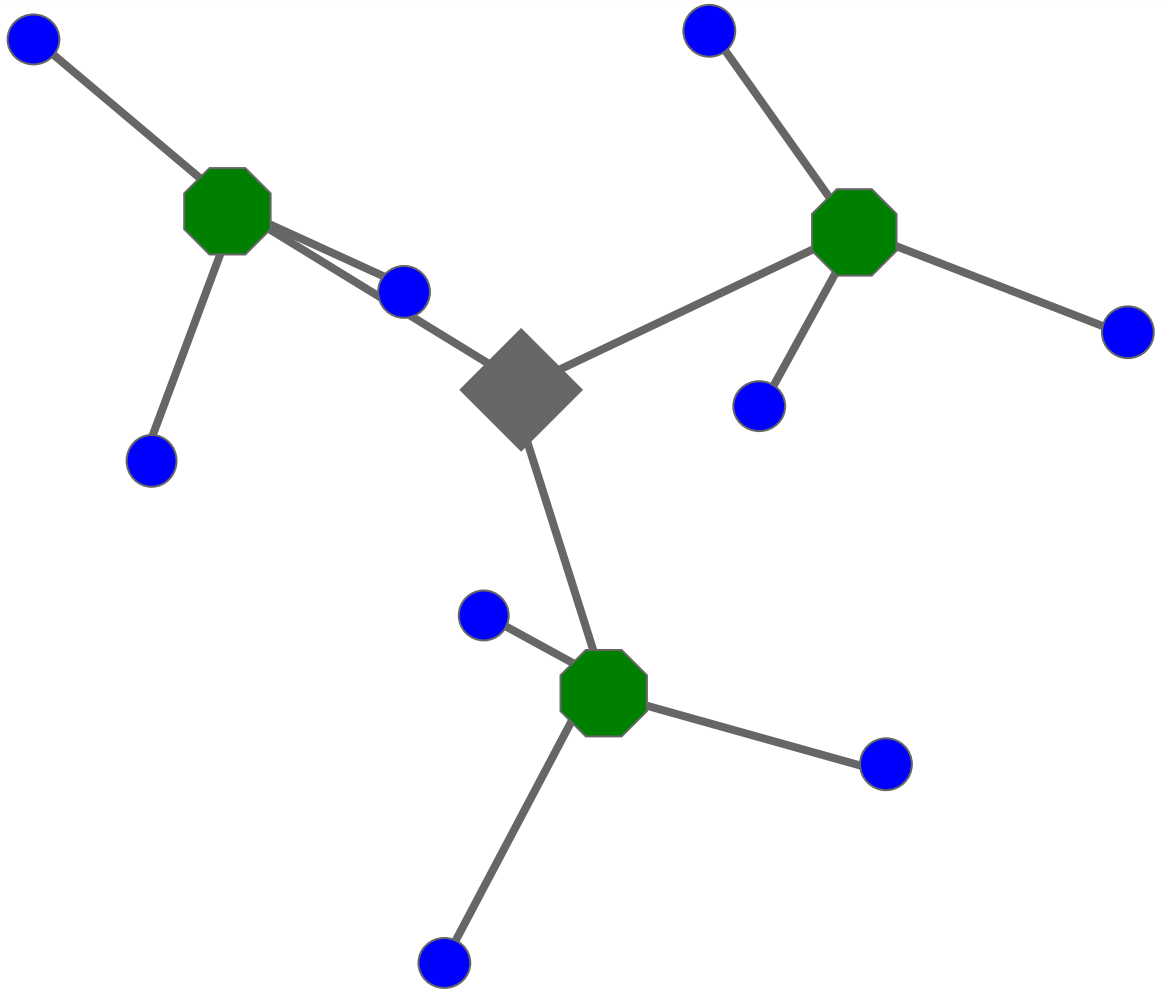
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



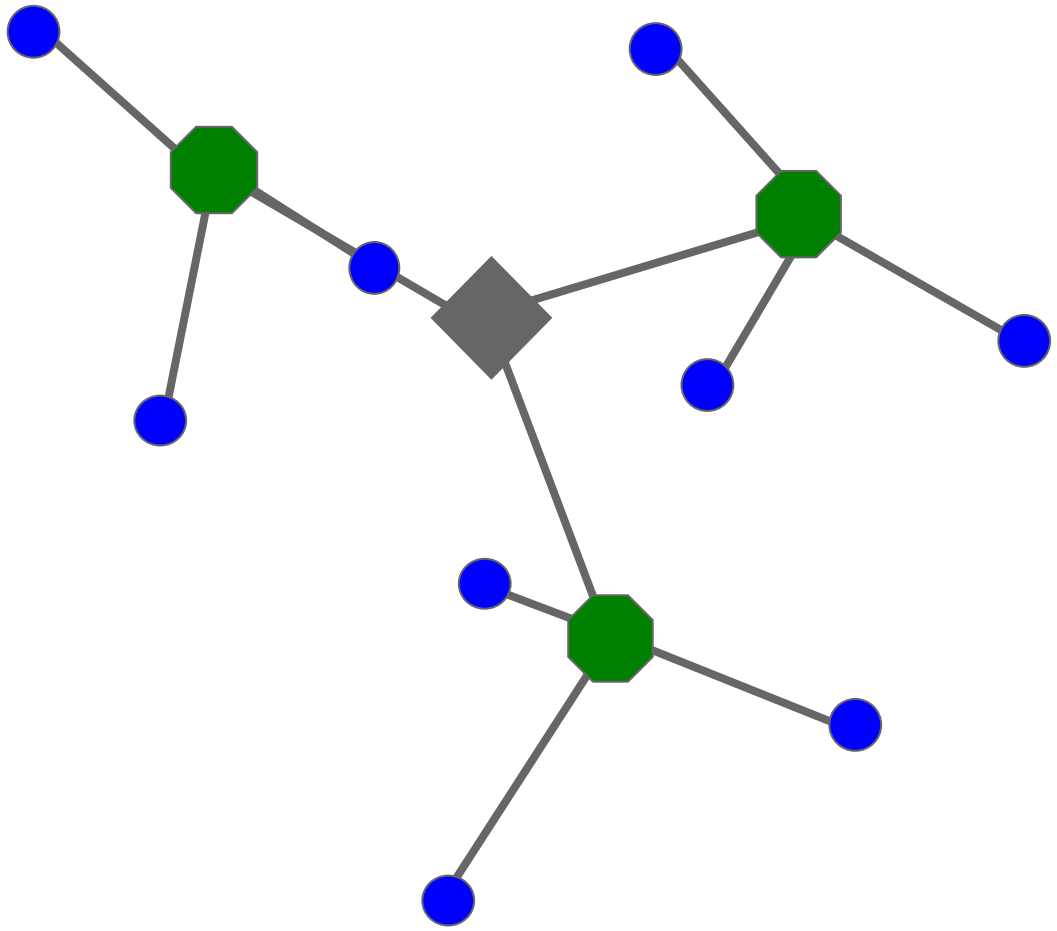
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



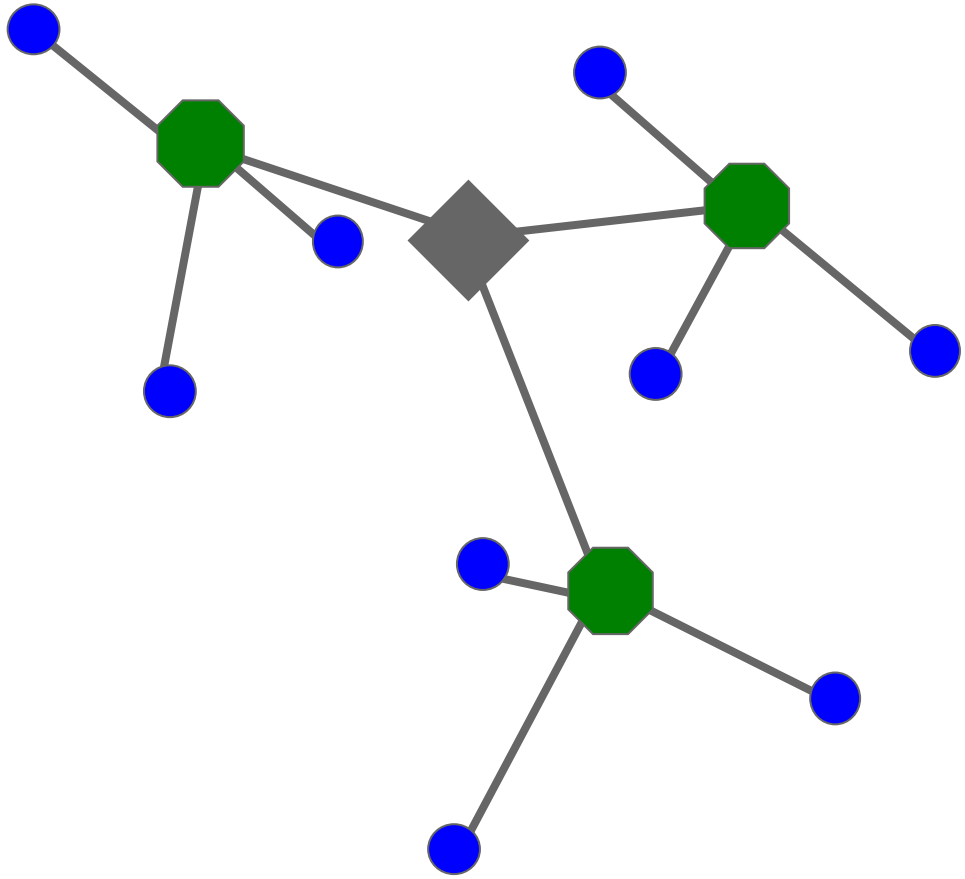
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



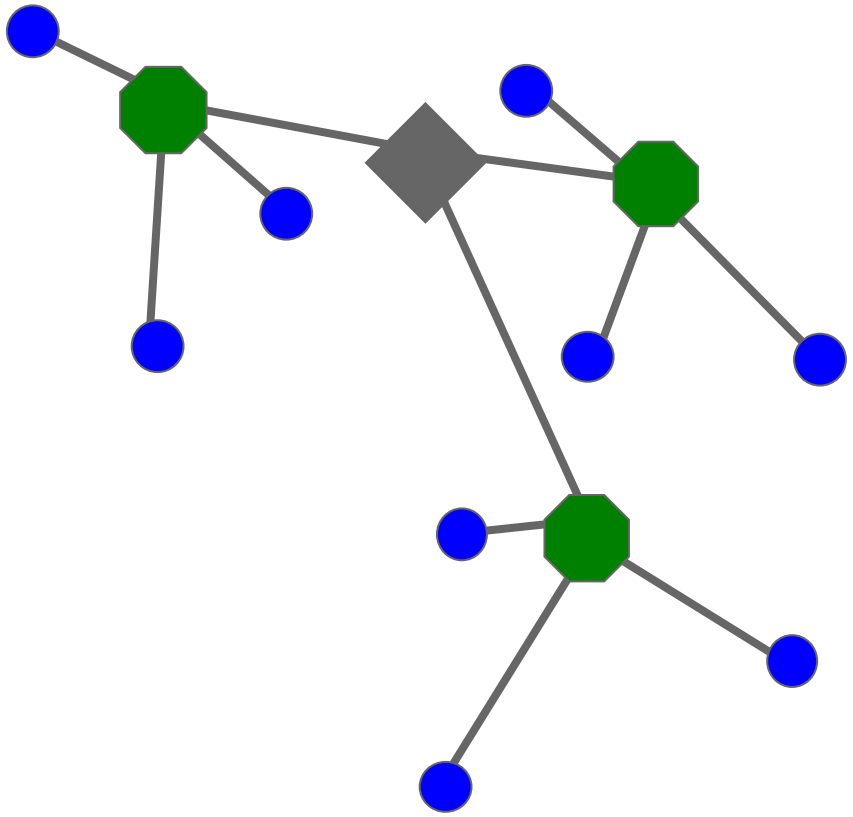
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



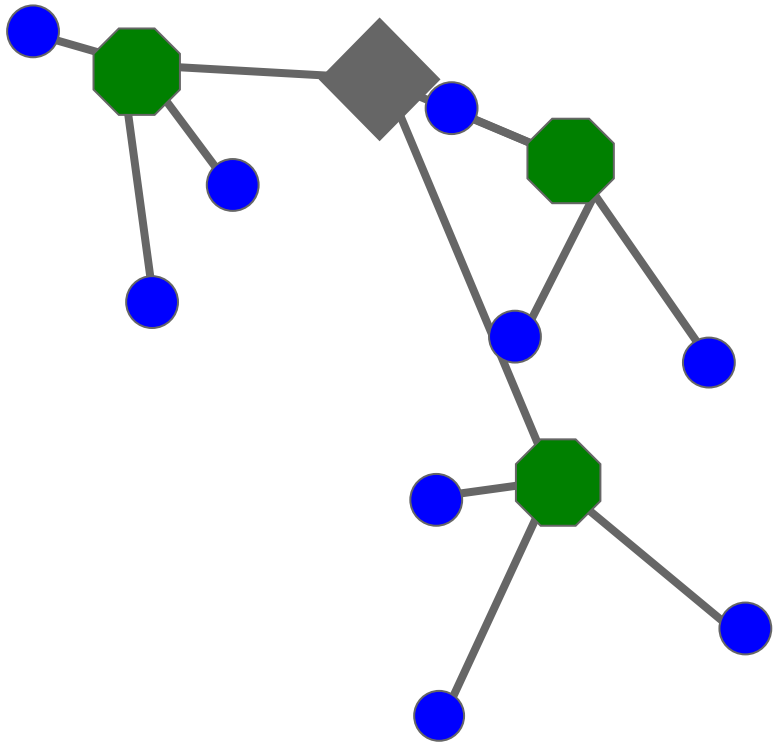
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



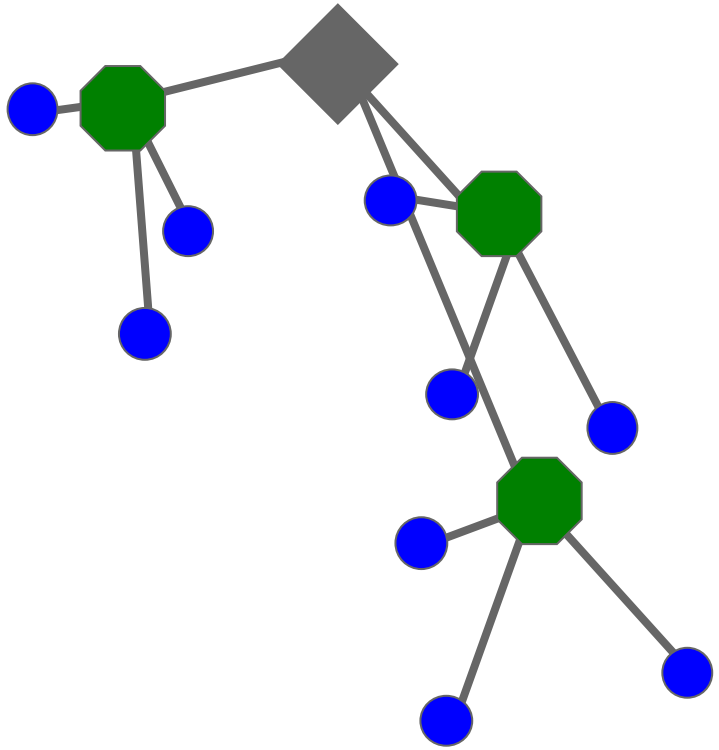
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



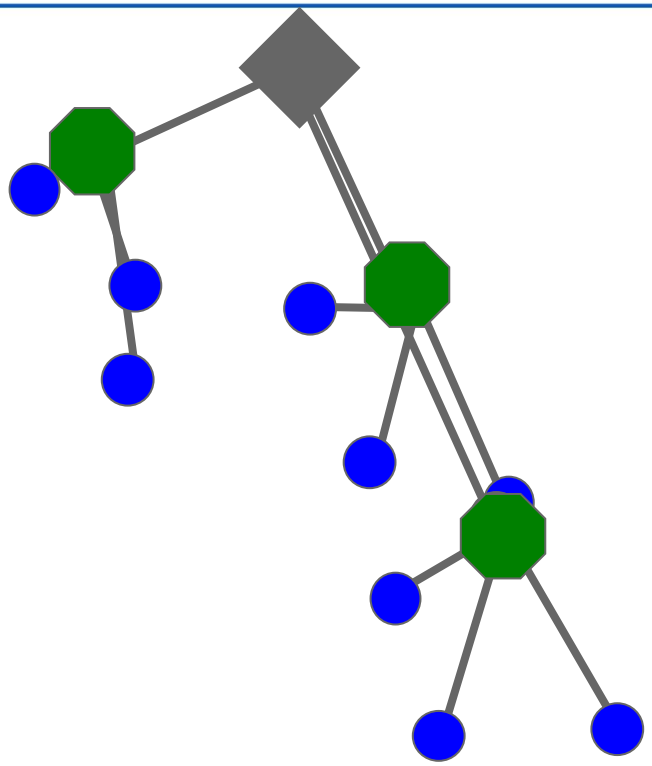
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



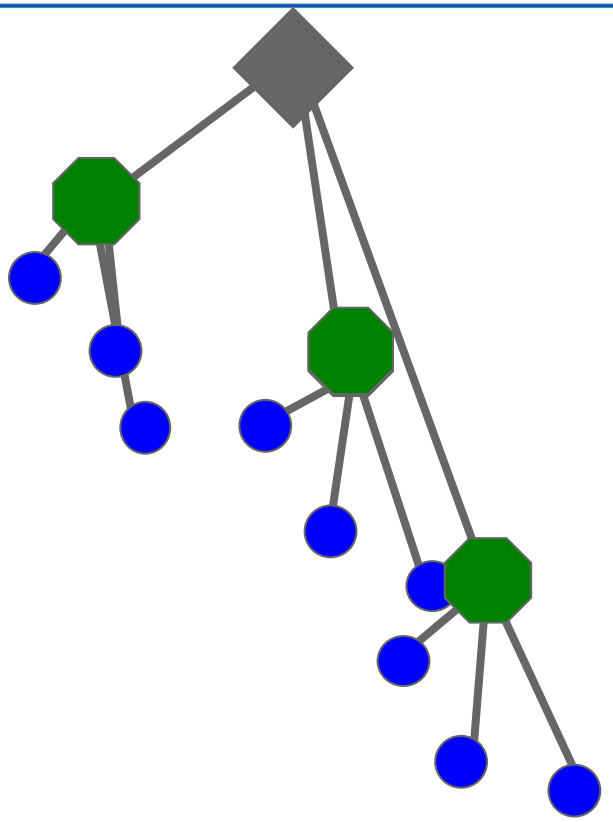
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



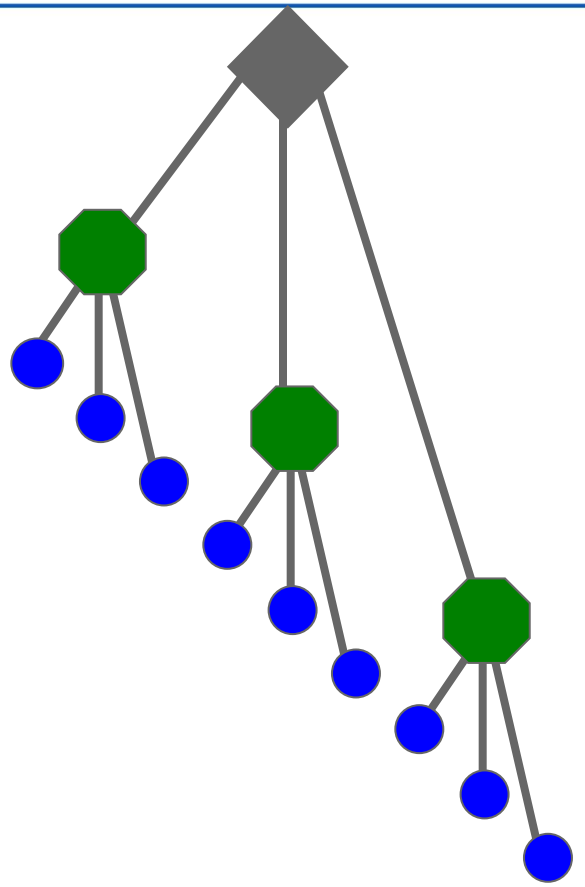
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



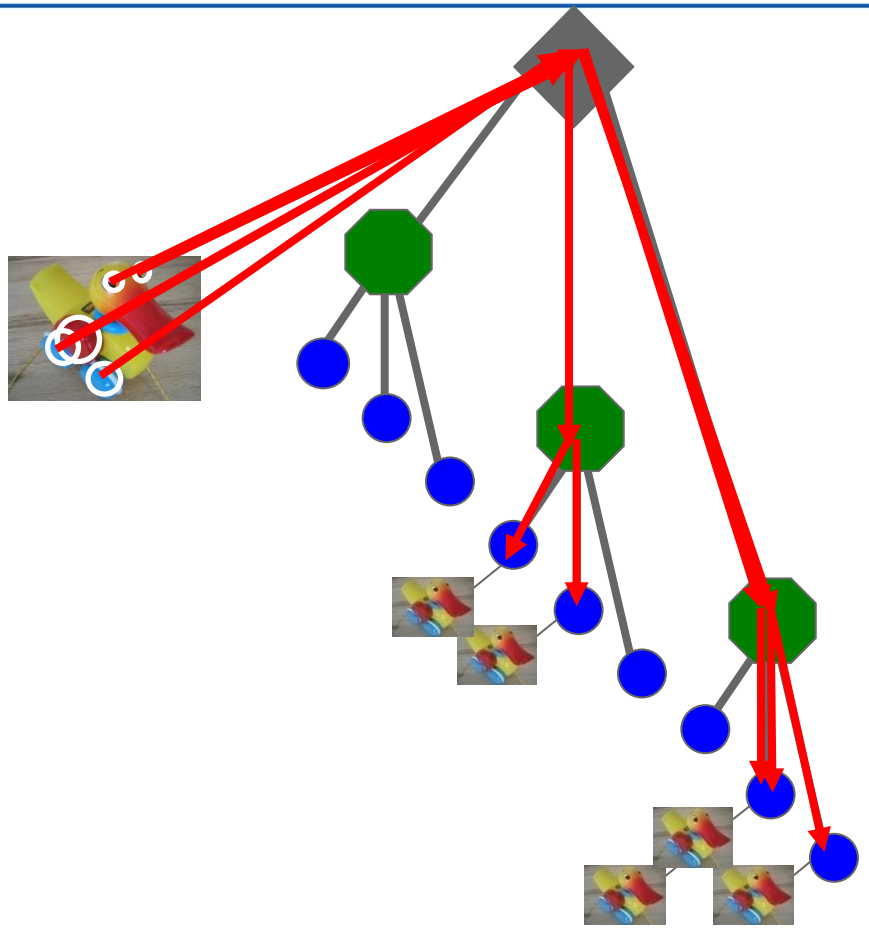
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



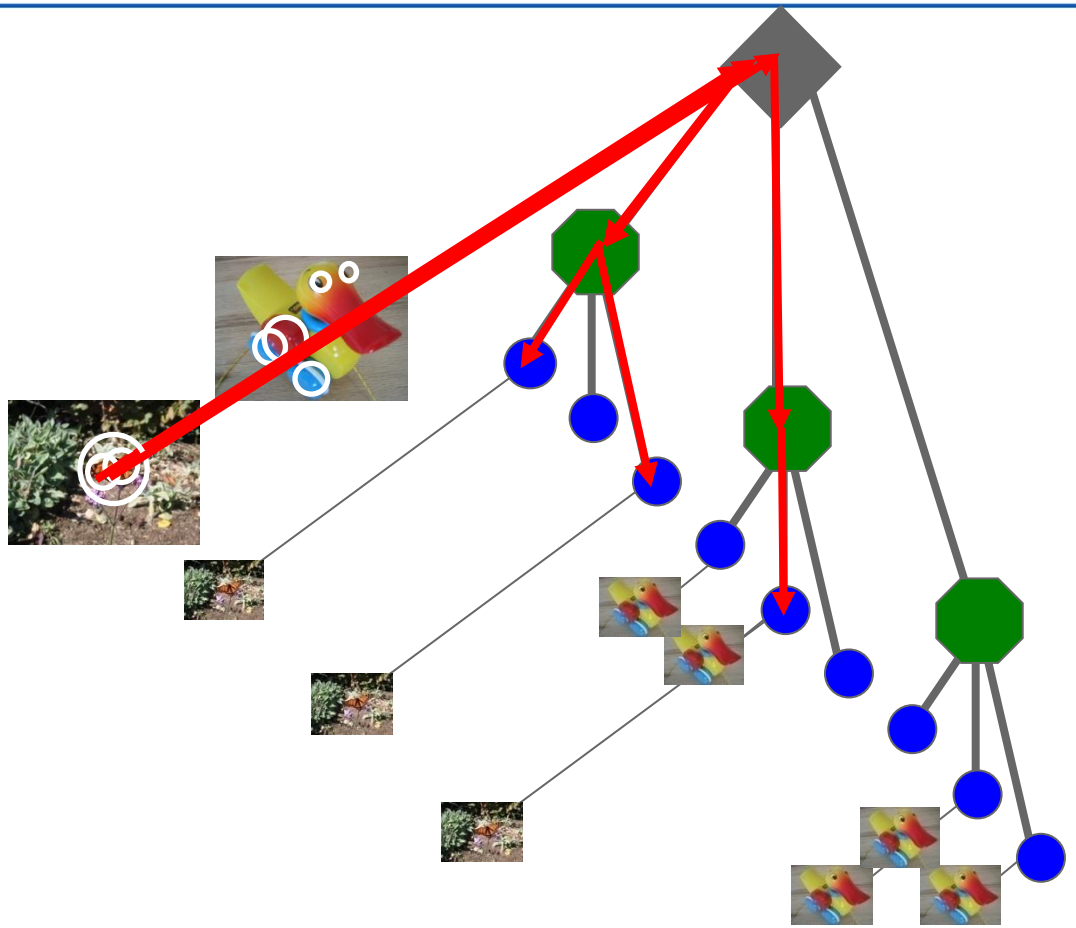
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



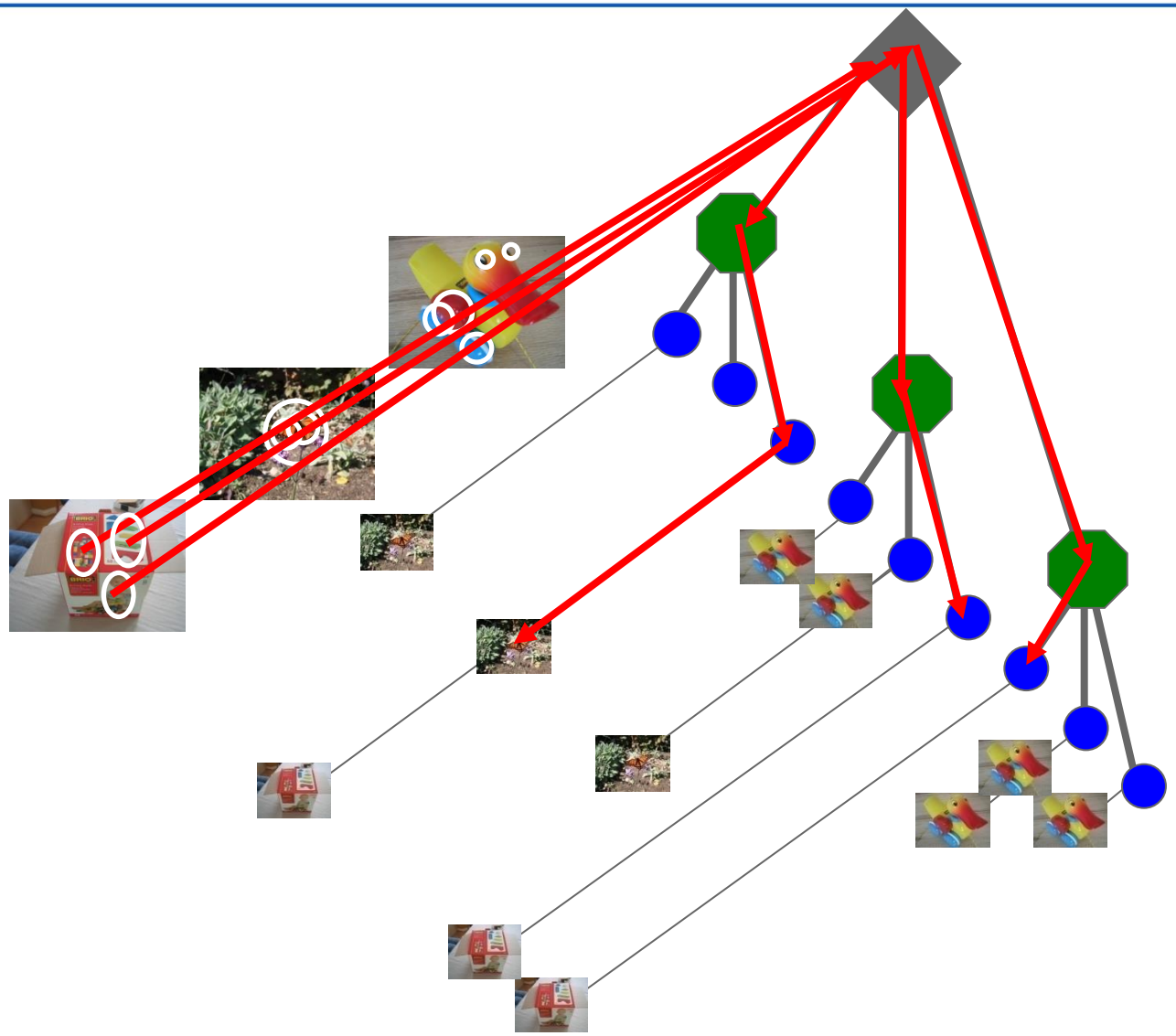
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



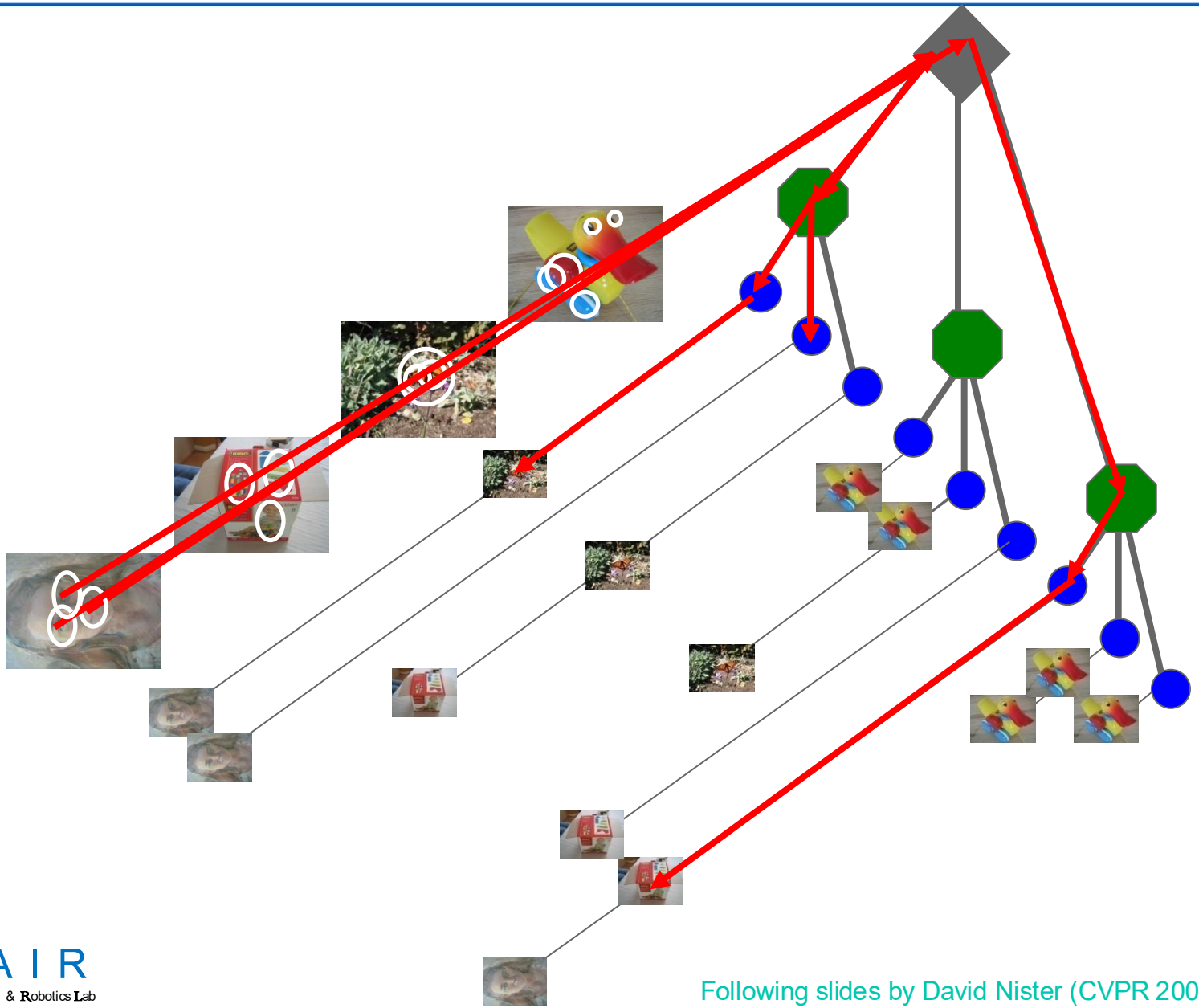
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



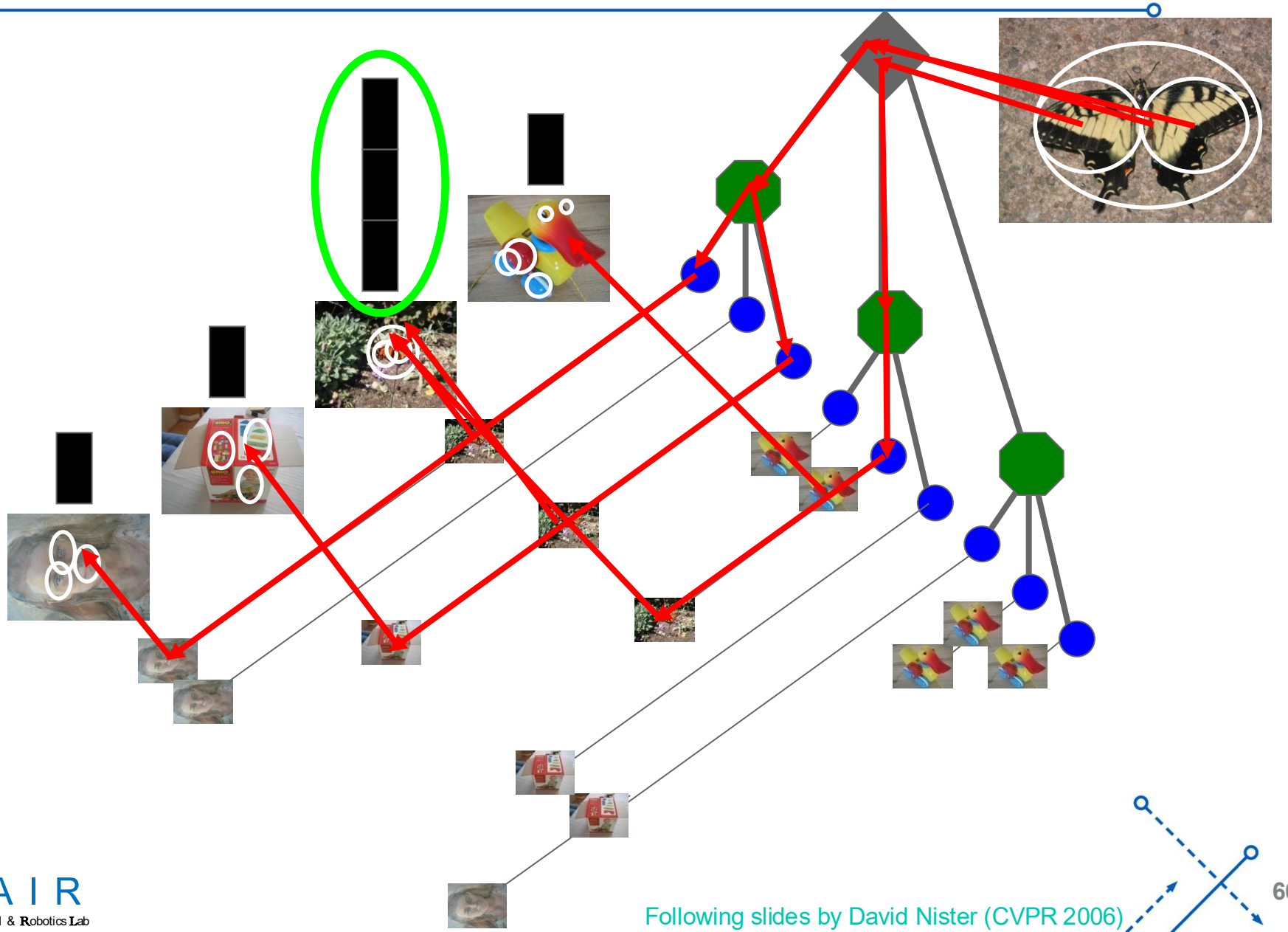
Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree



Following slides by David Nister (CVPR 2006)

Recognition with K-d Tree

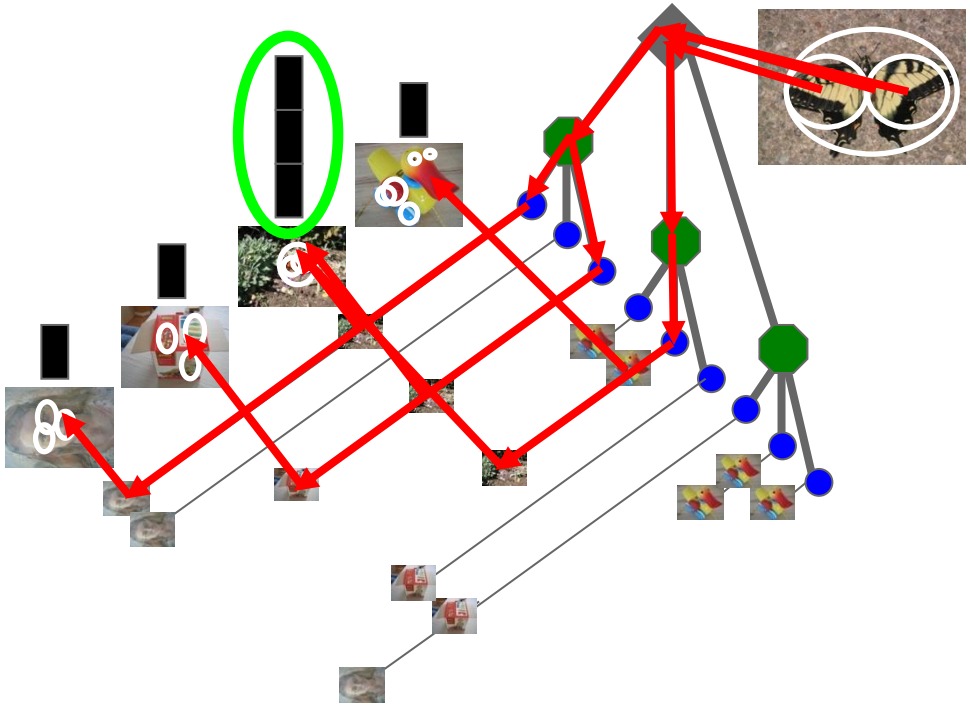


Following slides by David Nister (CVPR 2006)

Retrieval with K-d Tree

- Assume 1,000,000 visual words
  - Each node has 10 child nodes.
- How many comparison we need for a query feature?

$$\log_{10} 1,000,000 = 6$$



Vocabulary Tree: Performance

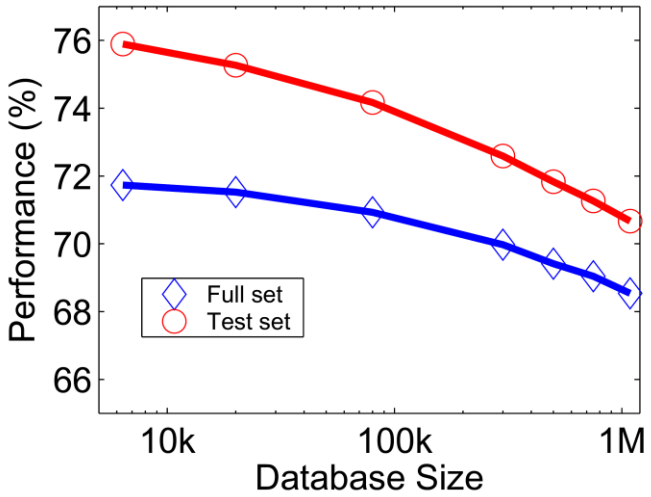
Evaluated on large databases

- Indexing with up to 1M images

Online recognition for database of 50,000 CD covers

- Retrieval in ~1s

Find experimentally that large vocabularies can be beneficial for recognition

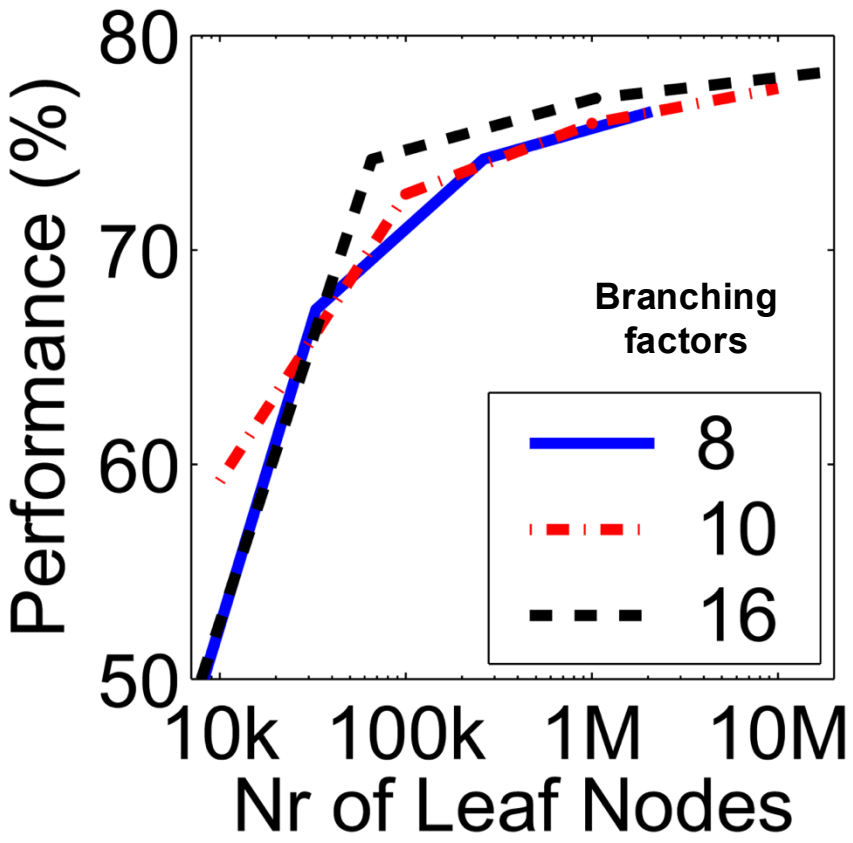




[Nister & Stewenius, CVPR'06]

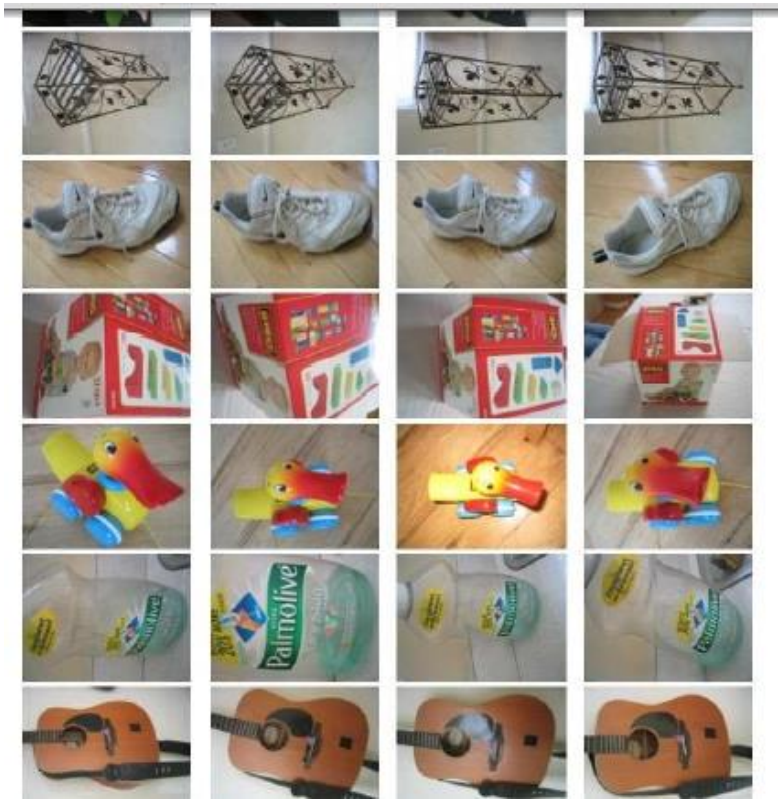
Slide credit: J. Sivic

How large should the vocabulary be?



Influence on performance, sparsity

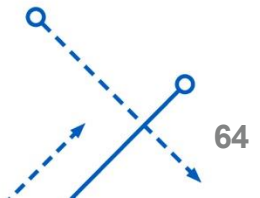
Results for recognition task with 6347 images



Nister & Stewenius, CVPR 2006

Visual vocabularies: Issues

- How to choose vocabulary size?
  - Too small
    - visual words not representative of all patches
  - Too large:
    - quantization artifacts, overfitting



(Nister & Stewenius, 2006)

Visual words/bags of words

Pro

+ flexible to geometry / deformations / viewpoint
+ compact summary of image content
+ provides fixed dimensional vector representation for sets
+ good results in practice

Cons

- background and foreground mixed when bag covers whole image
- optimal vocabulary formation remains unclear
- basic model ignores geometry
  - must verify afterwards, or encode via features

How to score the retrieval results?

- Precision, Recall, Area under PR curve



Precision, Recall, F1 (Recap)

How precise are the answers you gave?

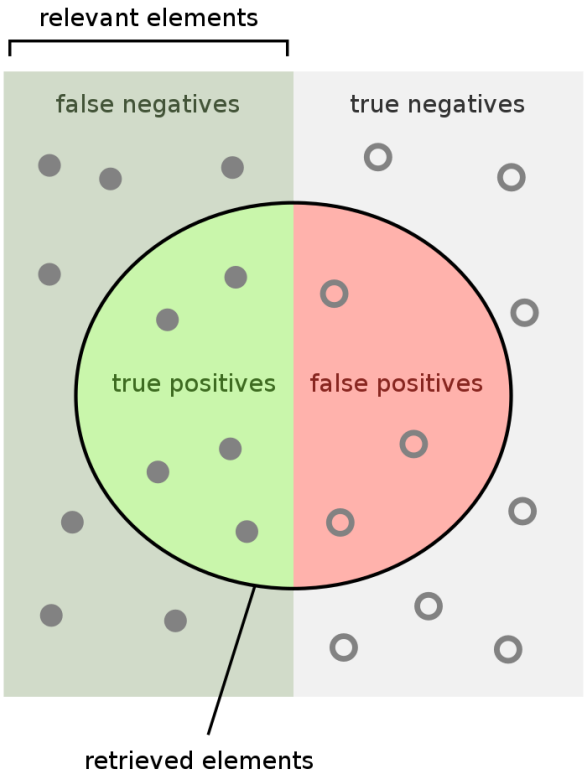
$$\text{Precision} = \frac{\text{True Positive}}{\text{Predicted Positive}} = \frac{\text{\# Relevant}}{\text{\# Total Returned}}$$

How many did you find of the ones that can be found?

$$\text{Recall} = \frac{\text{True Positive}}{\text{Actual Positive}} = \frac{\text{\# Relevant}}{\text{\# Total Revevant}}$$

F1: Harmonic Mean of Precision and Recall

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2\frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$



How many retrieved items are relevant?

Precision =

How many relevant items are retrieved?

Recall =

Precision-Recall Curve (PR Curve)

Database size: 10 images
Relevant (total): 5 images

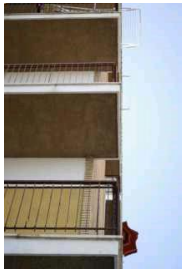
precision = #relevant / #returned
recall = #relevant / #total relevant

Query



Results (ordered):

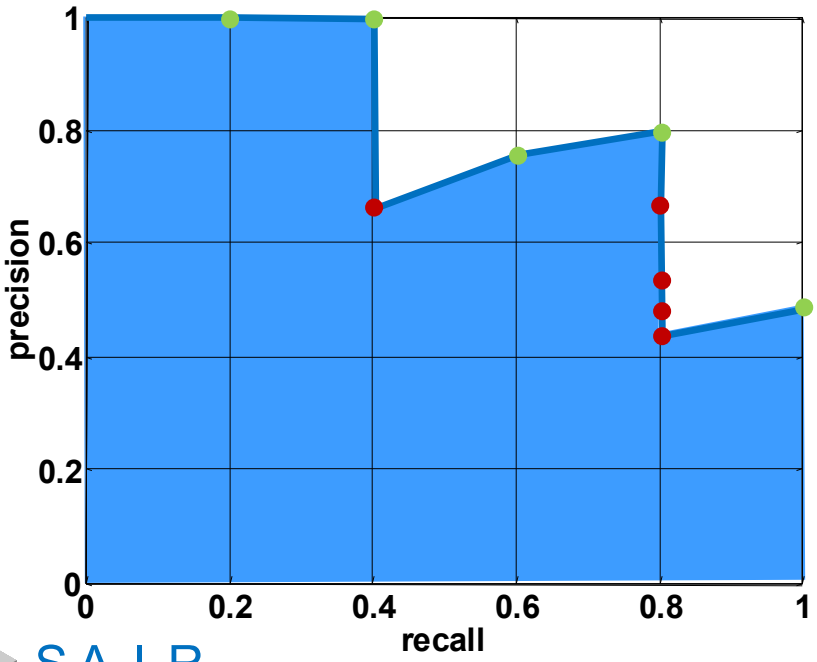










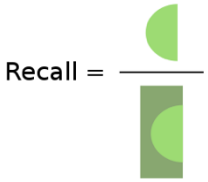
SAIR
Spatial AI & Robotics Lab

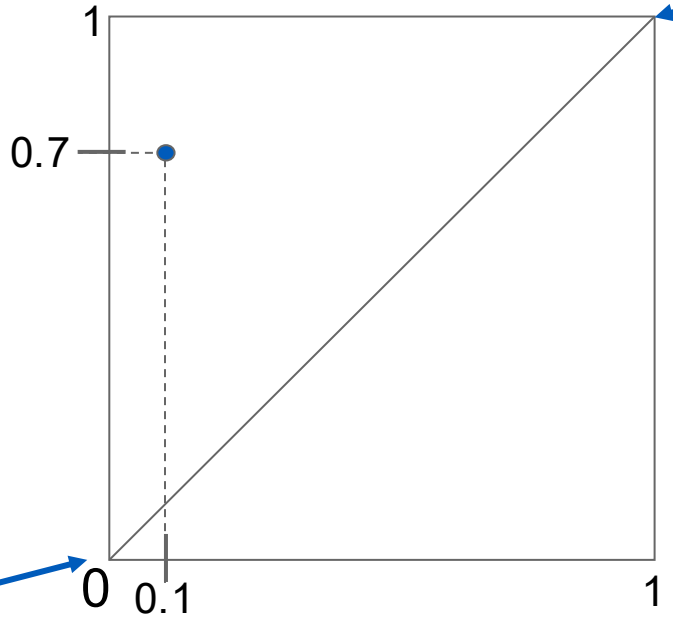
Slide credit: Ondrej Chum

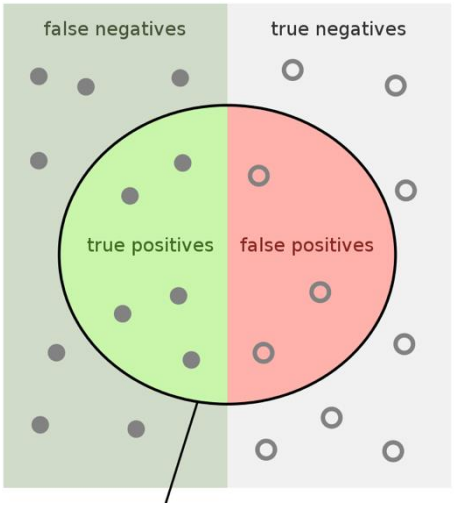
ROC curve (Recap)

- How can we measure the performance of a feature matcher?

Lowest threshold
all pairs as match







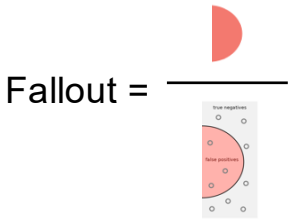
True Positive Rate

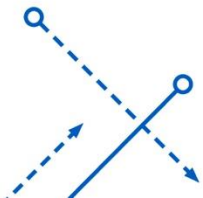
$$\frac{\#\,TP}{\#\,TP + FN}$$

Highest threshold
no match

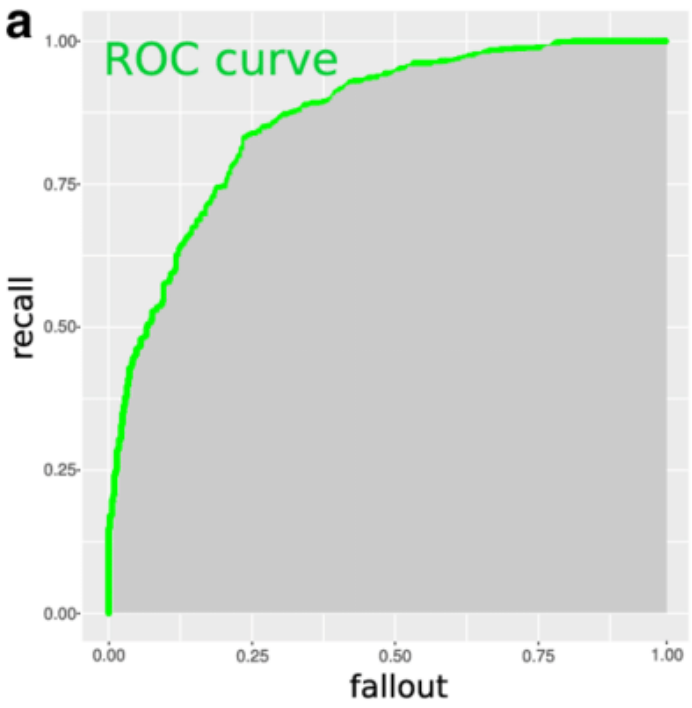
False Positive Rate

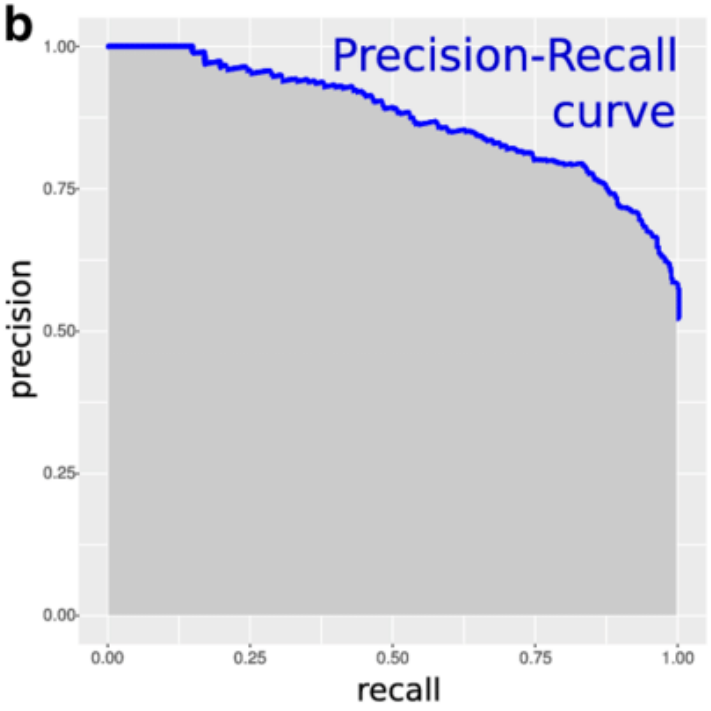
$$\frac{\#\,FP}{\#\,FP + TN}$$





ROC curve VS. PR curve





Recall = 

Precision = 

Fallout = 

Recall = 

ROC curve VS. PR curve

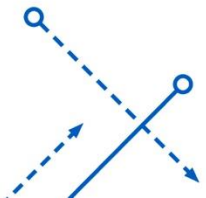
- ROC Curves (Lecture 4)
  - Summarize the trade-off between the **true positive rate** and **false positive rate**.

- PR curves (This Lecture)
  - Summarize the trade-off between the **true positive rate** and the **positive predictive value**.



ROC curve VS. PR curve

- ROC
  - Good for balanced datasets/classes.
- PR curves
  - Suitable for **imbalanced datasets**.
- If the ratio of positive to negative instances changes:
  - The ROC curves will not change
  - ROC curves do not depend on class distributions
- ROC “weights” equally true positives and true negatives.
  - Not expected sometimes, e.g., place recognition.



More explanation